

# **A birth–death–catastrophe extension of the standard within-host model**

Renewal dynamics forced by an intracellular process,  
and when bursting pays

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# Chapter 1

## A birth–death–catastrophe extension of the standard within-host model, and what it says about bursting

### 1.1 Introduction

The standard model of within-host virus dynamics describes an infected cell with two numbers: it releases virions at a constant rate  $p$  and is removed at a constant rate  $d_I$  [1, 2, 3]. Both are fitted to viral load data and then reported as though they characterised the pathogen. They are, however, phenomenological. Nothing in the standard model says what the cell is doing between infection and release, and the constants absorb whatever it is.

For one class of pathogen this costs nothing. A budding virus really does experience release and death as separate, independent events, and section 1.2.2 shows that the standard model is then an exact description of the cell, not an approximation to it. For a lytic pathogen it is wrong in mechanism. Such a pathogen releases nothing while it replicates inside the cell and then releases everything at once, in the event that ends the cell’s productive life. Release and death are not independent; they are the same event. *Y. pestis* in the macrophage, *F. tularensis* in the macrophage and the lytic phage in its bacterium are all of that kind, and none is a constant drip.

That the standard model needs intracellular structure is not a new observation, and the modelling programme built on it is reviewed by Ciupe and Conway [4]. Two routes are established. The first inserts a delay between infection and production, either as a distributed kernel or as one or more explicit eclipse compartments, an approach going back to Herz et al. [5] and now standard [6, 7, 8]. The second, multiscale, route couples an age-structured population equation to an explicit intracellular pool: in the hepatitis C models of Guedj et al. and Rong and Perelson [9, 10] an intracellular viral RNA level  $R(t, a)$  is tracked alongside infected cells  $I(t, a)$  of infection age  $a$ , and free virus is fed by  $\rho \int_0^\infty R(t, a) I(t, a) da$ . Both routes have paid off. The review’s two headline returns are sharper parameter estimates — neglecting the pre-productive period biases  $R_0$  [11] — and the ability to locate a drug’s mode of action at a named step of the viral life cycle.

What neither route supplies is a *stochastic* intracellular mechanism. The delay literature assumes a distributional family for the age structure and fits it; the multiscale literature derives the age structure, but from a deterministic pool, so it yields a release schedule and nothing more. Yet for a lytic pathogen the questions that matter early in infection are distributional. Whether a founder cell establishes an infection at all depends on how many particles it releases, not on how many it releases on average, and a cell whose internal population dies out before rupture releases none.

This chapter closes that gap for one mechanism. The chapter on the birth–death–catastrophe process and the chapter on its distribution theory developed a birth–death–catastrophe process for the intracellular replicative units — birth at rate  $\lambda$  per unit, death at rate  $\mu$  per unit, and catastrophe at rate  $\delta$  per unit, at which the cell ruptures and releases its entire contents — and obtained its survival, release and burst-size laws in closed form. Both stop at the cell boundary. Here that theory is carried across the boundary into a population.

## Contributions

1. **A renewal model forced by the intracellular process** (section 1.4). The population equations need exactly two functions of cell age: the survival kernel  $S$  and the release kernel  $g$ . Both are already in closed form, so the population model has no free age structure to fit. The equations are the first moments of the underlying stochastic process exactly, not an approximation to them (proposition 1.2).
2. **The standard model is a projection** (section 1.5). For each epidemic growth rate  $r$  there is a unique constant pair  $(p_{\text{eff}}(r), d_{\mathcal{L},\text{eff}}(r))$  reproducing the renewal system’s exponential solution exactly (theorem 1.3). A fitted release rate is therefore a property of the infection’s phase, not of the pathogen: over the whole map it moves by the factor  $\mathbb{E}[X^2]_{\text{QS}}$ , a pure number fixed by the intracellular rates, which is 231 at  $(\lambda, \mu, \delta) = (1, 0, 0.1)$ . No constant pair serves a transient, and no fit of  $(p, d_{\mathcal{L}})$  recovers the three intracellular rates (section 1.5.6).
3. **A closed-form answer to whether bursting pays** (section 1.6). This is the result the stochastic treatment exists to deliver, since it turns on the offspring *distribution* rather than its mean. Write  $L = a(1 - b)$  for the flooding parameter (definition 1.7). At fixed total output per cell, bursting leaves  $R_0$  exactly unchanged (theorem 1.6); lowers the extinction probability at establishment if and only if  $L > 1$ , equivalently

$$\frac{1}{\delta} < \frac{1}{\lambda} + \frac{1}{\mu},$$

by the exact amount  $(L - 1)(1/m - 1)$  (theorem 1.8); and slows deterministic invasion in every regime examined. Threshold, establishment risk and invasion speed are three questions with three different answers.

4. **Commensurability** (section 1.7). Because only the pair  $(S, g)$  enters, budding, bursting, recurrent partial release, multiply infected cells and the multiscale hepatitis C model of [9, 10] are all instances of one skeleton, differing only in the kernels they supply (table 1.4).

Table 1.7 collects every population-level formula of the chapter in one place, for use as a reference card. Throughout,  $\lambda, \mu$  and  $\delta$  are the intracellular birth, death and catastrophe rates per replicative unit; the roots  $a, b$ , the derived constants  $A, B, \theta, \kappa$  and the single-cell functions  $I, D, I_{\text{fix}}, J, K, V$  are those of the chapter on the birth–death–catastrophe process, collected in section 1.A. Table 1.1 fixes the population symbols. Two naming decisions are made once and kept: the burst-time density written  $\varphi$  in the chapter on its distribution theory is written  $\delta J$  here, because  $\varphi$  is reserved for the partial-release fraction of section 1.7.2; and the survival kernel  $S$  is the non-fixation probability  $I_{\text{fix}}$ , written  $S$  where it acts as a kernel and  $I_{\text{fix}}$  where the closed form is meant.

### 1.1.1 Standing assumptions

These hold throughout unless stated otherwise.

(A1) Target cells  $T$  are held constant: no depletion, no saturation, no immune response.

| Symbol                     | Meaning                                                                      | Notes                    |
|----------------------------|------------------------------------------------------------------------------|--------------------------|
| $\alpha$                   | cell age                                                                     | argument of every kernel |
| $T, \gamma, c$             | target cells, infection rate, virion clearance                               | $T$ held fixed           |
| $\mathcal{I}, \mathcal{V}$ | infected cells, free particles                                               | population counts        |
| $i(t)$                     | incidence, $\gamma T \mathcal{V}$                                            |                          |
| $d_{\mathcal{I}}$          | infected-cell removal rate                                                   | phenomenological         |
| $p, p_{\text{eff}}(r)$     | constant, effective release rate                                             |                          |
| $S(\alpha), g(\alpha)$     | survival kernel $I_{\text{fix}}(\alpha)$ , release kernel $\delta K(\alpha)$ |                          |
| $\mathcal{A}(\alpha)$      | generation kernel                                                            | (1.21)                   |
| $r$                        | epidemic growth rate                                                         |                          |
| $q, m$                     | infection success $\gamma T / (\gamma T + c)$ , branching mean $qV_{\infty}$ |                          |
| $L$                        | flooding parameter $a(1 - b)$                                                | (1.27)                   |
| $\varphi$                  | partial-release fraction                                                     | section 1.7.2            |

**Table 1.1:** Population notation. Single-cell symbols are those of section 1.A.

(A2) Virions are not consumed on infection — the *continuum* convention. This is why the deterministic threshold is  $R_0 = \gamma T V_{\infty} / c$  while the discrete branching mean of section 1.6 is  $m = qV_{\infty}$  with  $q = \gamma T / (\gamma T + c)$ ; the two agree when  $\gamma T \ll c$ .

(A3) The  $\mathcal{I}_0$  cells present at  $t = 0$  all have age exactly zero.

(A4) Infected cells are independent, and the population layer assumes single infection: kernels for founding multiplicity  $k$  are available (section 1.A) but are not used.

(A5) There is no spatial structure.

## 1.2 The standard model and the cell it describes exactly

### 1.2.1 The standard model

In the target-cell-constant Basic Model of Viral Replication (BMVR), infected cells  $\mathcal{I}$  and free particles  $\mathcal{V}$  obey

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad (1.1)$$

with target cells  $T$  fixed, infection rate  $\gamma$  and virion clearance  $c$ . The basic reproduction number is

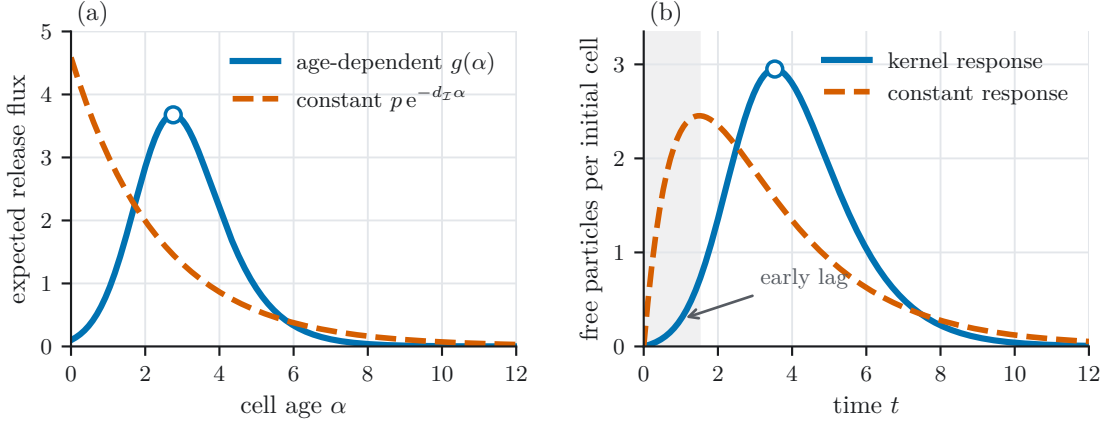
$$R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}. \quad (1.2)$$

Every cell, whatever its age or contents, releases at rate  $p$  and dies at rate  $d_{\mathcal{I}}$ .

### 1.2.2 The budding cell is described exactly

This is not an arbitrary caricature. It is one kind of cell described exactly, and since that cell is the comparator for everything below it is worth setting out in full. A budding infected cell experiences two *independent* event streams: release of one particle, as a Poisson process of rate  $p$ , and death, as a Poisson process of rate  $d_{\mathcal{I}}$ . Its survival and cumulative release are

$$S_{\text{bud}}(\alpha) = e^{-d_{\mathcal{I}}\alpha}, \quad V_{\text{bud}}(\alpha) = \frac{p}{d_{\mathcal{I}}} \left(1 - e^{-d_{\mathcal{I}}\alpha}\right), \quad (1.3)$$



**Figure 1.1:** A synchronous cohort releases on a schedule, not at a rate. Both schedules have lifetime yield  $V_\infty = 11$  and mean productive lifetime  $\mathbb{E}[T_{\text{prod}}] = 2.398$ , with  $d_I = 1/\mathbb{E}[T_{\text{prod}}]$ ,  $p = V_\infty d_I$ ,  $(\lambda, \mu, \delta) = (1, 0, 0.1)$  and  $c = 1$ : as fairly matched as two constants allow. (a) At matched yield the release kernel  $g(\alpha) = \delta K(\alpha)$  still lags the budding cell's  $p e^{-d_I \alpha}$ , because a bursting cell must first accumulate load. (b) Linear clearance preserves that lag, carrying it into the free-particle response.

so the mean productive lifetime is  $1/d_I$  and the lifetime release  $p/d_I$ . The total number released is geometric on  $\{0, 1, 2, \dots\}$ ,

$$\mathbb{P}\{\mathcal{K}_{\text{bud}} = k\} = \frac{d_I}{d_I + p} \left( \frac{p}{d_I + p} \right)^k, \quad (1.4)$$

the successes before the terminating death in a sequence of events each of which is a release with probability  $p/(d_I + p)$ . All of the cell sits in the pair  $(p, d_I)$ . That is why (1.1) is a faithful one-cell limit for a budding pathogen, and it is this cell that section 1.6 matches against a bursting one.

### 1.2.3 Bursting, and why no constant repairs it

In bursting, release and death are the same event: whatever has accumulated leaves at once and the cell's productive life ends in that instant. Take the cleanest consequence. Let  $N_0$  cells be infected at the same moment, each running an independent copy of the single-cell process. The number still productively infected at time  $t$  is  $N_0 I_{\text{fix}}(t)$  exactly in expectation, and the expected cumulative release is  $N_0 V(t)$ , so the release *rate* is  $N_0 \delta K(t)$ . It depends on the age of the cohort and it is not constant. A cohort releases on a schedule, not at a rate (figure 1.1).

The natural repair is to keep (1.1) and rebuild  $p$  from the intracellular theory — to set  $p$  equal to the quasi-stationary mean load  $\mathbb{E}[X]_{\text{QS}} = a/(a - 1)$ , on the reasoning that a long-lived cell hovers at that level. The obstruction is dimensional. Here  $p$  is a number of particles per unit time and  $\mathbb{E}[X]_{\text{QS}}$  is a pure count, so the substitution introduces a spurious factor of time; when  $\mu = 0$  one has  $\mathbb{E}[X]_{\text{QS}} = V_\infty$  exactly, and the proposal replaces a cell's output *rate* by its total lifetime *output*. Averaging the burst size over the burst-time density fails the same way, returning  $\delta \int_0^\infty K = V_\infty$ , a count again. Weighting the load-dependent flux by the burst-time density has the right dimensions but the wrong measure: that density says when a *single* cell bursts, whereas a population release rate needs an average over the ages of the cells *currently infected*, and in an asynchronous infection those are not the same.

What survives is two facts rather than a substitute constant. The expected release flux of a cell of age  $\alpha$  is  $\delta K(\alpha)$ ; and at any host time the infected population is a mixture of ages, one cohort per past moment of infection. The population model must therefore track cell age.



## 1.3 The intracellular process and its two kernels

### 1.3.1 The process

Inside a cell, replicative units follow a linear birth–death process with an added catastrophe. Each unit gives birth at rate  $\lambda$ , dies at rate  $\mu$ , and at rate  $\delta$  per unit triggers rupture, at which the cell releases its entire load and enters an absorbing state. Writing  $X_\alpha$  for the load at cell age  $\alpha$ , the process is absorbed either by rupture — a burst — or by internal extinction, in which the last unit dies and the cell never bursts at all. The chapter on the birth–death–catastrophe process solves it. The roots

$$a = \eta + \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad \eta = \frac{\lambda + \mu + \delta}{2\lambda}, \quad (1.5)$$

satisfy  $b < 1 < a$  and carry everything:  $b$  is the probability that a cell clears internally without ever bursting, and  $1/a$  is the ratio of the geometric burst-size law. [Section 1.A](#) collects the closed forms used below; the two that matter most are the lifetime yield and the mean productive lifetime,

$$V_\infty = \frac{a(1-b)}{a-1}, \quad \mathbb{E}[T_{\text{prod}}] = \frac{1}{\lambda} \log \frac{a}{a-1}. \quad (1.6)$$

### 1.3.2 The two kernels

Age-structured population dynamics needs exactly two single-cell ingredients. The *survival kernel* is the probability that a cell infected  $\alpha$  time units ago is still productively infected,

$$S(\alpha) := I_{\text{fix}}(\alpha) = \frac{(a-b)^2 e^{\theta\alpha}}{(B + Ae^{\theta\alpha})(ae^{\theta\alpha} - b)}, \quad (1.7)$$

and the *release kernel* is the expected release flux at cell age  $\alpha$ ,

$$g(\alpha) := \delta K(\alpha) = V'(\alpha). \quad (1.8)$$

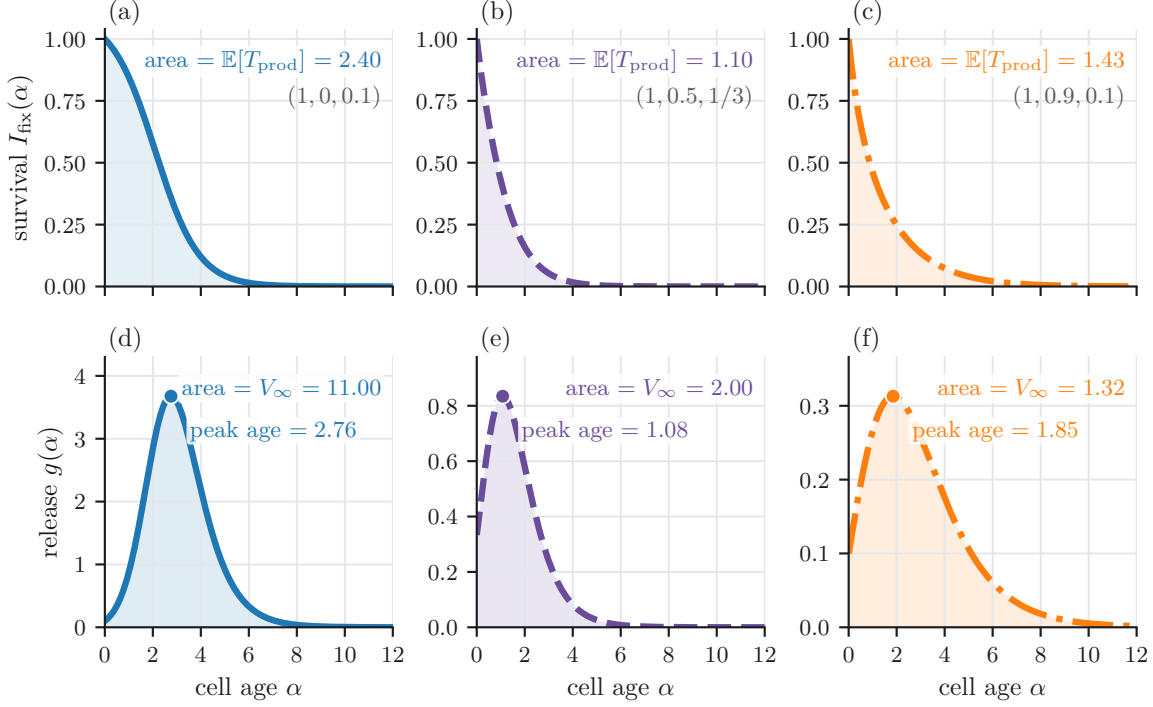
The reasoning behind (1.8) is the one the failed proposal of [section 1.2.3](#) stated and then did not use: a cell bursts at rate  $\delta X_\alpha$  and releases  $X_\alpha$  particles when it does, so the expected number released in  $[\alpha, \alpha + d\alpha]$  is  $\delta \mathbb{E}[X_\alpha^2] d\alpha = \delta K(\alpha) d\alpha$ . With  $K$  from (1.43) both kernels are explicit functions of age ([figure 1.2](#)). This is what it will mean for the population model to be *forced by* the intracellular process: the kernels are computed, not fitted.

### 1.3.3 Three regimes, used throughout

Three intracellular parameter triples recur below, chosen to straddle the flooding boundary  $L = 1$  of [section 1.6](#): one on each side, one exactly on it. They are named here and used by name thereafter.

| $(\lambda, \mu, \delta)$ | $a$   | $b$   | $L$  | $V_\infty$ | $\mathbb{E}[T_{\text{prod}}]$ | $d_{\mathcal{I}, \text{eff}}(0)$ | $p_{\text{eff}}(0)$ |
|--------------------------|-------|-------|------|------------|-------------------------------|----------------------------------|---------------------|
| (1, 0, 0.1)              | 1.100 | 0     | 1.10 | 11.00      | 2.398                         | 0.417                            | 4.59                |
| (1, 0.5, 1/3)            | 1.500 | 1/3   | 1.00 | 2.00       | 1.099                         | 0.910                            | 1.82                |
| (1, 0.9, 0.1)            | 1.316 | 0.684 | 0.42 | 1.32       | 1.426                         | 0.701                            | 0.92                |

**Table 1.2:** The three regimes used throughout. The first has  $L > 1$ , the second sits on the boundary  $L = 1$ , the third has  $L < 1$ . The last two columns are the  $r = 0$  endpoints of the effective-parameter map of [section 1.5](#).



**Figure 1.2:** Survival  $I_{\text{fix}}$  (a)–(c) and release  $g = \delta K$  (d)–(f) across the flooding boundary of [table 1.2](#). Release rises then falls — a cell must accumulate before it bursts — while survival decays; shaded areas check  $\int_0^\infty I_{\text{fix}} = \mathbb{E}[T_{\text{prod}}]$  and  $\int_0^\infty g = V_\infty$ . Parameters, left to right:  $(\lambda, \mu, \delta) = (1, 0, 0.1), (1, 0.5, 1/3), (1, 0.9, 0.1)$ , with  $L = 1.10, 1.00, 0.42$  and  $V_\infty = 11.00, 2.00, 1.32$ . Independent  $g$ -scales keep the timing shift visible.

## 1.4 Burst-aware renewal dynamics

### 1.4.1 The renewal system

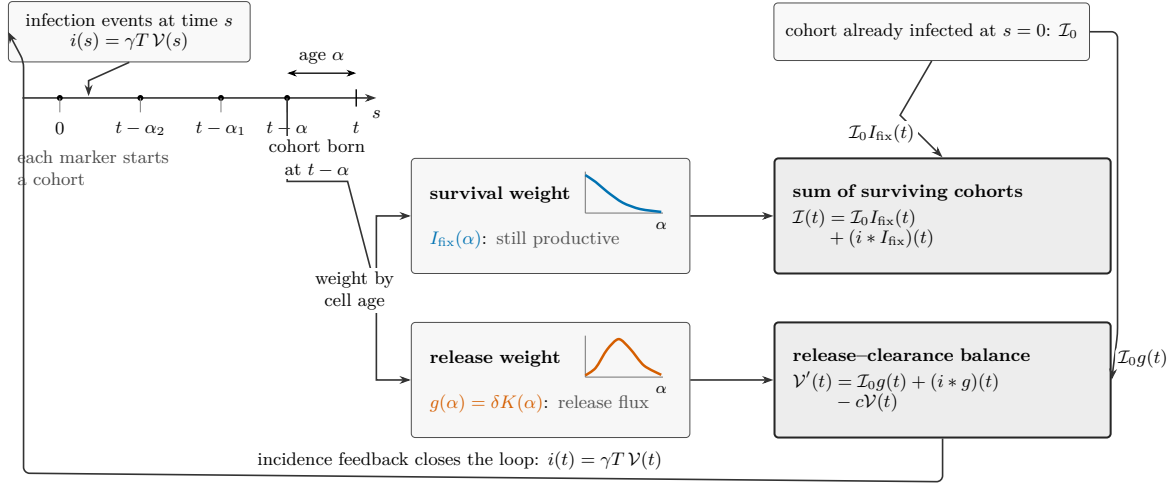
Hold target cells  $T$  fixed, let infection occur at rate  $\gamma$  per virion, clear virions at rate  $c$ , and define the incidence  $i(t) = \gamma T \mathcal{V}(t)$ . Each infection event at time  $t - \alpha$  starts a cell of age  $\alpha$ , still productive with probability  $S(\alpha)$  and releasing at expected rate  $g(\alpha)$ . Summing over infection times and adding the  $\mathcal{I}_0$  cells already infected at  $t = 0$  gives the system, whose bookkeeping is drawn in [figure 1.3](#).

**Definition 1.1** (The burst-aware renewal system).

$$\mathcal{I}(t) = \mathcal{I}_0 I_{\text{fix}}(t) + \int_0^t i(t - \alpha) I_{\text{fix}}(\alpha) d\alpha = \mathcal{I}_0 I_{\text{fix}} + (i * I_{\text{fix}})(t), \quad (1.9)$$

$$\frac{d\mathcal{V}}{dt}(t) = \mathcal{I}_0 g(t) + \int_0^t i(t - \alpha) g(\alpha) d\alpha - c \mathcal{V}(t) = \mathcal{I}_0 g + (i * g)(t) - c \mathcal{V}(t). \quad (1.10)$$

[Table 1.3](#) sets the two systems side by side. Both constants are replaced, not only the release: memoryless removal  $d_{\mathcal{I}} \mathcal{I}$  becomes the age structure implicit in [\(1.9\)](#), and constant release  $p \mathcal{I}$  becomes the convolution with  $g$ . A replacement touching only the release term — the first thought of [section 1.2.3](#) — would leave the removal side memoryless, and still wrong. The removal side is not merely a formal concern: analysing stage-resolved HIV decay under drugs acting at different points in the life cycle, Gilmore et al. [\[12\]](#) found infected-cell half-lives close to those of uninfected cells before integration and much shorter after it, which is an age-dependent  $d_{\mathcal{I}}$  measured directly.



**Figure 1.3:** Bookkeeping of [definition 1.1](#): infection events start cohorts; the kernels  $I_{\text{fix}}$  and  $g = \delta K$  weight them by age; the feedback  $i = \gamma T V$  closes the loop.

| Standard BMVR (1.1)                              | Renewal system (1.9)–(1.10)                                       |
|--------------------------------------------------|-------------------------------------------------------------------|
| $\mathcal{I}' = i - d_{\mathcal{I}} \mathcal{I}$ | $\mathcal{I} = \mathcal{I}_0 I_{\text{fix}} + i * I_{\text{fix}}$ |
| constant removal rate $d_{\mathcal{I}}$          | age-structured survival via $I_{\text{fix}}$                      |
| $\mathcal{V}' = p \mathcal{I} - c \mathcal{V}$   | $\mathcal{V}' = \mathcal{I}_0 g + i * g - c \mathcal{V}$          |
| constant release rate $p$                        | age-structured release via $g = \delta K$                         |

**Table 1.3:** What the renewal system replaces. Both constants go, not only the release rate.

Differentiating (1.9) makes the structure visible:

$$\frac{d\mathcal{I}}{dt} = i(t) + \mathcal{I}_0 I'_{\text{fix}}(t) + \int_0^t i(t - \alpha) I'_{\text{fix}}(\alpha) d\alpha,$$

in which incidence appears as a source exactly as in the standard model, while the remaining terms encode age-dependent loss of productivity. There is in general no constant  $d_{\mathcal{I}}$  reducing this to  $\mathcal{I}' = i - d_{\mathcal{I}} \mathcal{I}$  for every history of  $i$ ; that holds only when everything grows exponentially, and the exception is what [section 1.5](#) exploits.

### 1.4.2 The equations are exact in the mean

Equations (1.9) and (1.10) are not an approximation to the underlying stochastic process. They are its first moments exactly, and the reason is that every rate in sight is linear.

**Proposition 1.2 (First-moment exactness).** *Consider the stochastic process in which each infection event starts an independent birth–death–catastrophe cell, each released particle independently infects at rate  $\gamma T$  or clears at rate  $c$ , and the  $\mathcal{I}_0$  cells present at  $t = 0$  have age zero. Write  $\mathcal{I}(t)$  and  $\mathcal{V}(t)$  for the expected numbers of productive cells and free particles. Then  $\mathcal{I}$  and  $\mathcal{V}$  satisfy (1.9) and (1.10) exactly.*

*Proof.* Let  $N$  be the point process of infection events. Since each free particle infects at rate  $\gamma T$  and particles are independent,  $N$  has compensator  $\gamma T \mathcal{V}^\omega(t) dt$ , where  $\mathcal{V}^\omega$  is the realised particle count; taking expectations, its mean intensity is  $i(t) = \gamma T \mathcal{V}(t)$ . A cell founded at time  $t - \alpha$  evolves independently of  $N$  and of every other cell, so it is still productive at  $t$  with probability  $I_{\text{fix}}(\alpha)$  and its expected release flux at  $t$  is  $g(\alpha)$ , both by definition of the kernels. Conditioning on  $N$  and applying Campbell's

formula — legitimate because  $I_{\text{fix}}$  and  $g$  are bounded and non-negative, so Fubini applies — gives  $\mathcal{I}(t) = \mathcal{I}_0 I_{\text{fix}}(t) + (i * I_{\text{fix}})(t)$  and an expected release flux of  $\mathcal{I}_0 g(t) + (i * g)(t)$ . Clearance removes particles at rate  $c$  per particle, contributing  $-c\mathcal{V}(t)$  to the mean. Collecting terms gives (1.10).  $\square$

The step that would fail for a nonlinear incidence is the identification of the mean intensity with  $\gamma T\mathcal{V}$ : for  $f(\mathcal{V}, T)$  nonlinear,  $\mathbb{E}[f(\mathcal{V}^\omega)] \neq f(\mathbb{E}[\mathcal{V}^\omega])$  and the moment hierarchy does not close. Assumption (A1) is what keeps the incidence linear. This is one reason to prefer mass action where the data permit it, and the review of [4] reports that they generally do, the saturating alternatives — Holling type 2, Beddington–DeAngelis, Crowley–Martin — being well explored theoretically but not validated against data.

### 1.4.3 One skeleton, many mechanisms

Nothing in (1.9) and (1.10) used the birth–death–catastrophe process beyond the two kernels it supplied. With incidence  $i(t) = \gamma T\mathcal{V}(t)$ , or a nonlinear  $f(\mathcal{V}, T)$ , *any* pair  $(S, g)$  gives

$$\mathcal{I}(t) = \mathcal{I}_0 S(t) + (i * S)(t), \quad \frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g(t) + (i * g)(t) - c\mathcal{V}(t), \quad (1.11)$$

with effective parameters  $p_{\text{eff}}(r) = \tilde{g}(r)/\tilde{S}(r)$  and, for linear incidence,  $R_0 = (\gamma T/c) \int_0^\infty g$ . This is the chapter’s most general statement, and it is what makes release mechanisms commensurable: they differ only in  $(S, g)$  (table 1.4). The first row is a structural check on the whole construction: exponential kernels reduce (1.11) exactly to standard BMVR (section 1.4.5).

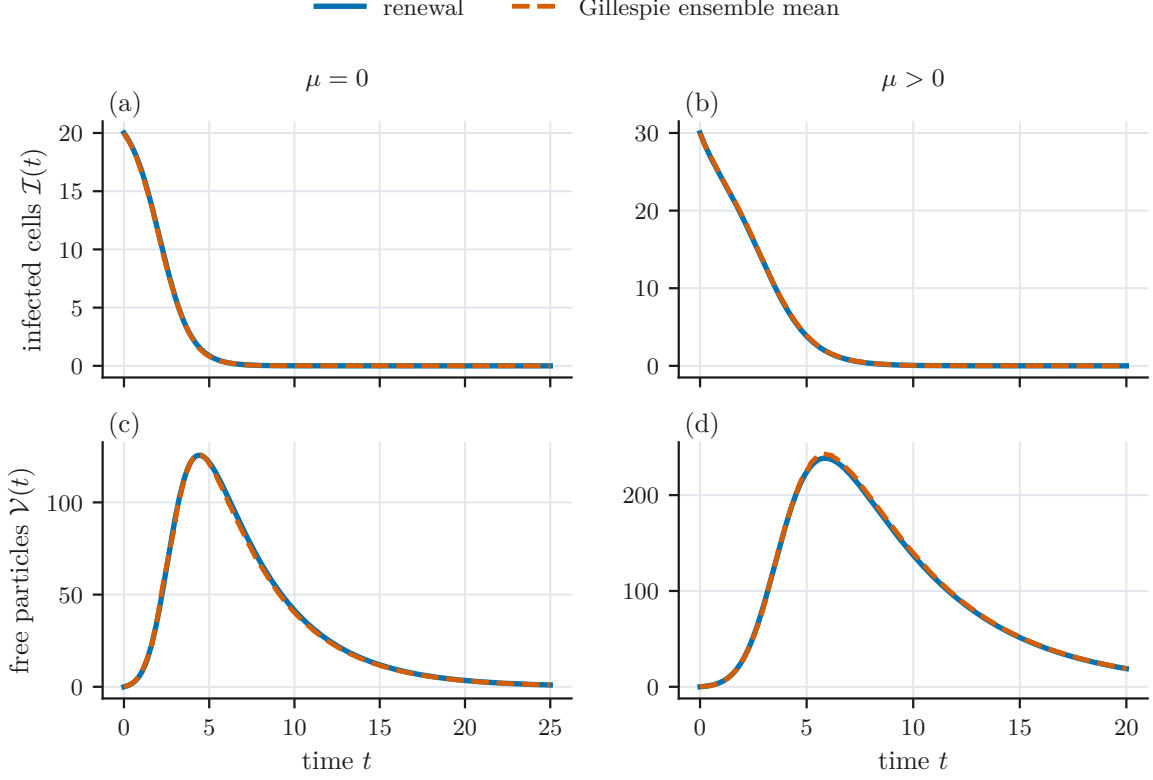
| Model                                                 | $S(\alpha)$                 | $g(\alpha)$                                                              |
|-------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------|
| Standard BMVR [3]                                     | $e^{-d_I \alpha}$           | $p e^{-d_I \alpha}$                                                      |
| Eclipse + standard [4]                                | $e^{-d_I \alpha}$           | $p e^{-d_I \alpha} \mathbf{1}\{\alpha > \tau_e\}$                        |
| HCV multiscale [9, 10]                                | $e^{-\delta \alpha}$        | $\rho R(\alpha) e^{-\delta \alpha}$ , with $R$ from an intracellular ODE |
| Absorbing BDC (this chapter)                          | $I_{\text{fix}}(\alpha)$    | $\delta K(\alpha)$                                                       |
| MOI- $k$ BDC (the chapter on its distribution theory) | $I(\alpha)^k - D(\alpha)^k$ | $\delta K_k(\alpha)$                                                     |
| Reset catastrophe (section 1.7.1)                     | $e^{-d_I \alpha}$           | $\delta e^{-d_I \alpha} \mathbb{E}[X_\alpha^2]$                          |
| Two-type / GATE (the two-type chapter)                | two-type $I_{\text{fix}}$   | two-type second-moment flux                                              |

**Table 1.4:** One skeleton, many mechanisms. Each row is a different single-cell story; all induce (1.11), and differ only in the pair of kernels they supply.

### 1.4.4 Relation to existing multiscale models

The renewal framework itself is standard: it is the McKendrick–von Foerster machinery of mathematical demography [13, 14, 15]. So are age-structured within-host models. Nelson et al. [16] write down a within-host system with age-dependent production and death rates that is structurally (1.11), and take the age dependence as given. The multiscale hepatitis C models [9, 10] go further and derive it, coupling infected cells of age  $a$  to an intracellular viral RNA level  $R(t, a)$  obeying a transport equation with production, degradation and export; free virus is then fed by  $\rho \int_0^\infty R(t, a) I(t, a) da$ , which is (1.11) with  $g(\alpha) = \rho R(\alpha) e^{-\delta \alpha}$ . That model is the third row of table 1.4 and a direct ancestor of what is done here.

The difference is what the intracellular layer is. In [9, 10] it is a deterministic pool, so it supplies a mean release schedule and nothing else. Here it is a stochastic process solved in



**Figure 1.4:** Stochastic ground truth. Ensemble means of an independent Gillespie simulation — birth–death–catastrophe cells with impulse releases at burst times, exponential clearance, no reinfection — against (1.9)–(1.10). Left column  $\mu = 0$ , right column  $\mu > 0$ ; (a)–(b) infected cells, (c)–(d) free particles. The curves are indistinguishable at plotting resolution and the relative  $L^2$  discrepancy of 1–2% is sampling error: the renewal equations are not an approximation to the simulation, and this is the check that says so.

closed form, so it supplies the schedule *and* the distributions built on it: the burst-size law, the probability  $b$  that a cell clears internally and releases nothing, the offspring generating function, and hence extinction probabilities. Everything in section 1.5 could in principle be asked of a deterministic intracellular model; nothing in section 1.6 could, because establishment is a question about the offspring distribution rather than its mean. That is the sense in which the stochastic mechanism earns its place.

### 1.4.5 Numerical verification

The renewal system was checked end to end at four levels: kernel integrity, the reduction at  $\gamma T = 0$ , the structural limits available in closed form, and stochastic ground truth. All 62 checks pass. Exponential kernels reproduce standard BMVR to relative error below  $5 \times 10^{-5}$ ; the late growth rate equals the characteristic root of section 1.5 to about  $10^{-7}$ ; the  $R_0$  threshold is sharp; and an independent Gillespie ensemble reproduces the renewal means for  $\mathcal{I}$  and  $\mathcal{V}$  to within 1–2% (figure 1.4). The suite runs over the three regimes of table 1.2 together with  $(1, 0.2, 0.05)$  and  $(1, 0, 0.5)$ , chosen to spread the kernel shapes; it is the script `verify_result_20_1.py` distributed with the chapter source, and tests M and N of it are the two numerical sweeps cited in sections 1.5.3 and 1.6.4.

## 1.5 The standard model is a projection

The renewal system does not merely disagree with standard BMVR in transients. It contains standard BMVR as an exact special case, and the relationship is precise enough to be turned into predictions.

### 1.5.1 Exponential-phase equivalence

Suppose the incidence grows exponentially,  $i(t) = i_0 e^{rt}$ , at times late enough that the lower limits of the convolutions are irrelevant. Then

$$\mathcal{I}(t) = i_0 e^{rt} \widetilde{I_{\text{fix}}}(r), \quad \mathcal{V}'(t) = i_0 e^{rt} \delta \widetilde{K}(r) - c \mathcal{V}(t), \quad (1.12)$$

where  $\widetilde{f}(r) = \int_0^\infty e^{-r\alpha} f(\alpha) d\alpha$ . A standard model driven by the same incidence has  $\mathcal{I}(t) = i_0 e^{rt} / (r + d_{\mathcal{I}})$  and the eigenvalue relation  $(r + d_{\mathcal{I}})(r + c) = \gamma T p$ ; matching the two exponential solutions fixes

$$p_{\text{eff}}(r) = \frac{\delta \widetilde{K}(r)}{\widetilde{I_{\text{fix}}}(r)}, \quad d_{\mathcal{I},\text{eff}}(r) = \frac{1}{\widetilde{I_{\text{fix}}}(r)} - r. \quad (1.13)$$

**Theorem 1.3 (Exponential-phase equivalence).** *Fix a growth rate  $r > -\theta$ . The constant pair  $(p_{\text{eff}}(r), d_{\mathcal{I},\text{eff}}(r))$  of (1.13) is the unique one for which standard BMVR reproduces the renewal system's  $e^{rt}$  solution exactly, in both  $\mathcal{I}$  and  $\mathcal{V}$ ; and substituting it into the standard eigenvalue relation returns*

$$r + c = \gamma T \delta \widetilde{K}(r), \quad (1.14)$$

*precisely the characteristic equation of the renewal system. What no constant pair reproduces is a transient, which samples a continuum of growth rates at once.*

*Proof.* Uniqueness and existence are (1.12) read as two equations in two unknowns: matching  $\mathcal{I}$  forces  $1/(r + d_{\mathcal{I}}) = \widetilde{I_{\text{fix}}}(r)$ , which is  $d_{\mathcal{I},\text{eff}}$ , and matching the release term then forces  $p \widetilde{I_{\text{fix}}}(r) = \delta \widetilde{K}(r)$ , which is  $p_{\text{eff}}$ . Both transforms are finite and strictly positive for  $r > -\theta$ , so the pair is well defined. Substituting into  $(r + d_{\mathcal{I}})(r + c) = \gamma T p$  gives  $(r + c) \widetilde{I_{\text{fix}}}(r) = \gamma T \delta \widetilde{K}(r) / \widetilde{I_{\text{fix}}}(r)$ , that is (1.14). That no single pair serves a transient follows from proposition 1.5:  $p_{\text{eff}}$  is not constant in  $r$ .  $\square$

So the standard model is not a rival to the renewal system but a projection of it, exact on one exponential phase at a time. Everything else in this section reads (1.13) as a map.

### 1.5.2 The ends of the map

Using  $\int_0^\infty \delta K = V_\infty$  and  $\int_0^\infty I_{\text{fix}} = \mathbb{E}[T_{\text{prod}}]$ ,

$$r = 0 : \quad p_{\text{eff}}(0) = \frac{V_\infty}{\mathbb{E}[T_{\text{prod}}]} = \frac{\lambda V_\infty}{\log(a/(a-1))}, \quad d_{\mathcal{I},\text{eff}}(0) = \frac{\lambda}{\log(a/(a-1))}, \quad (1.15)$$

lifetime output divided by lifetime — a consistency check against naive moment-matching, and the matching used in the overlays below. At the other end,

$$r \rightarrow \infty : \quad p_{\text{eff}}(r) \rightarrow \delta, \quad d_{\mathcal{I},\text{eff}}(r) \rightarrow \mu + \delta, \quad (1.16)$$

the *young-cell* limit: in a fast-growing epidemic almost every infected cell was infected moments ago, holds one particle, bursts at rate  $\delta$  and releases one.

Calling these the two ends understates the map, because  $r = 0$  is not an end. Both transforms converge for every  $r > -\theta$ , and negative  $r$  is not a fiction — it is a decaying infection, the case of a treated patient. The genuine left-hand endpoint mirrors the young-cell limit.

**Proposition 1.4** (The old-cell limit). *The release flux per surviving cell tends to a finite limit,*

$$\lim_{\alpha \rightarrow \infty} \frac{g(\alpha)}{S(\alpha)} = \frac{a[2\lambda(1-b) + \delta]}{a-1} = \frac{\delta a(a+1)}{(a-1)^2} = \delta \mathbb{E}[X^2]_{\text{QS}}, \quad (1.17)$$

*the burst rate times the second moment of the quasi-stationary load, and consequently  $p_{\text{eff}}(r) \rightarrow \delta \mathbb{E}[X^2]_{\text{QS}}$  as  $r \downarrow -\theta$ .*

*Proof.* Write  $v = e^{-\theta\alpha}$ . As  $\alpha \rightarrow \infty$ ,  $v \rightarrow 0$  and (1.41) and (1.42) give  $I_{\text{fix}} \sim (a-b)^2 v / (Aa)$  and  $I \rightarrow b$ , so by (1.43)  $g = 2\lambda(\kappa - I)J \sim 2\lambda(\kappa - b)(a-b)^2 v / A^2$ . The ratio of the two is  $2\lambda(\kappa - b)a/A$ , which is the first equality after  $2\lambda(\kappa - b) = 2\lambda(1-b) + \delta$ . For the second, the identity  $AB = \delta/\lambda$  reads  $\lambda(1-b) = \delta/(a-1)$ , whence  $a[2\delta/(a-1) + \delta]/(a-1) = \delta a(a+1)/(a-1)^2$ . That last expression is  $\delta \mathbb{E}[X^2]_{\text{QS}}$  because the quasi-stationary load is geometric on  $\{1, 2, \dots\}$  with ratio  $1/a$  ((1.46)), whose second moment is  $a(a+1)/(a-1)^2$ . Finally, both  $\widetilde{I}_{\text{fix}}$  and  $\delta \widetilde{K}$  diverge like  $1/(r + \theta)$  as  $r \downarrow -\theta$ , with coefficients in the ratio just computed, so their quotient tends to it.  $\square$

The limit (1.17) also repairs the failed proposal of section 1.2.3, which offered the quasi-stationary *mean* as a release rate and was wrong twice over: a count where a rate was needed, and the first moment where the second is called for. Both corrections appear together —  $\delta$  supplies the missing inverse time, and the load-proportional burst hazard weights by  $X$  a second time.

### 1.5.3 Monotonicity, and the size of the effect

**Proposition 1.5** (Monotonicity of the effective release rate). *Write  $\rho(\alpha) = g(\alpha)/S(\alpha)$  for the expected release flux per still-productive cell, so that  $\rho(0) = \delta$  and  $\rho(\infty) = \delta \mathbb{E}[X^2]_{\text{QS}}$ . If  $\rho$  is non-decreasing then  $p_{\text{eff}}$  is non-increasing on  $(-\theta, \infty)$ , strictly so if  $\rho$  is strictly increasing.*

*Proof.* For  $r > -\theta$  let  $\pi_{f,r}$  be the probability density  $\alpha \mapsto e^{-r\alpha} f(\alpha)/\widetilde{f}(r)$ . Differentiating under the integral,  $d \log \widetilde{f}(r)/dr = -\mathbb{E}[\alpha]_{\pi_{f,r}}$ , so

$$\frac{d}{dr} \log p_{\text{eff}}(r) = \mathbb{E}[\alpha]_{\pi_{S,r}} - \mathbb{E}[\alpha]_{\pi_{g,r}}. \quad (1.18)$$

The two densities have ratio  $\pi_{g,r}(\alpha)/\pi_{S,r}(\alpha) \propto \rho(\alpha)$ , independently of  $r$ . If  $\rho$  is non-decreasing,  $\pi_{g,r}$  dominates  $\pi_{S,r}$  in the monotone likelihood ratio order, hence in the usual stochastic order, hence in mean; so (1.18) is non-positive.  $\square$

The hypothesis is a statement about one closed-form function of one variable: substituting  $v = e^{-\theta\alpha}$  gives

$$\rho = \frac{2\lambda[B(\kappa - a)v + A(\kappa - b)](a - bv)}{(Bv + A)^2}, \quad (1.19)$$

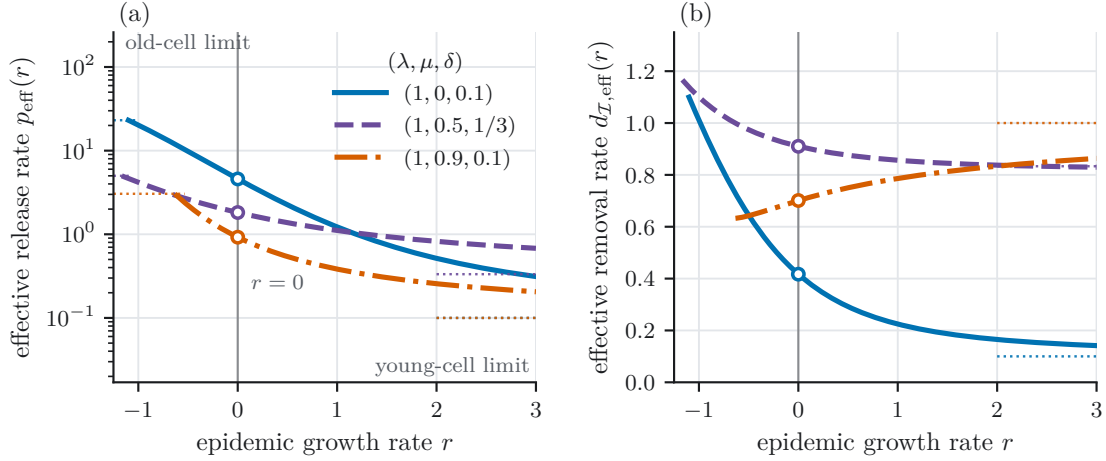
which must be shown decreasing in  $v \in (0, 1]$ . That reduces the question to a finite algebraic one. It is not settled here, but  $\rho$  was found increasing at every one of 4000 triples sampled over  $(\mu, \delta) \in [10^{-3}, 20]^2$  at  $\lambda = 1$ , and  $p_{\text{eff}}$  decreasing at every one (test M of the verification suite).

The size of the effect is what matters. At  $(\lambda, \mu, \delta) = (1, 0, 0.1)$ ,  $p_{\text{eff}}(0) \approx 4.59$  against  $p_{\text{eff}}(\infty) = 0.1$ : the same cell, two orders of magnitude apart, with nothing separating the two figures but how fast the epidemic around it is growing. Over the whole map the span is larger, and exact — by (1.16) and (1.17) the ratio of the endpoints is

$$\frac{\delta \mathbb{E}[X^2]_{\text{QS}}}{\delta} = \mathbb{E}[X^2]_{\text{QS}} = \frac{a(a+1)}{(a-1)^2}, \quad (1.20)$$

a pure number fixed by the intracellular rates alone. It is 231 at  $(1, 0, 0.1)$  and 30.5 at  $(1, 0.9, 0.1)$ : *the full dynamic range of a fitted release rate is the second moment of the quasi-stationary load.*





**Figure 1.5:** The effective release rate (a) and removal rate (b) of (1.13) against the epidemic growth rate  $r$ , for three parameter sets, over the whole domain  $r > -\theta$  on which the map is defined. Open circles mark the  $r = 0$  values (1.15); dotted segments mark the limits each curve runs toward. Note that  $r = 0$  is an interior point: the left-hand endpoint is the old-cell limit  $\delta \mathbb{E}_{\text{QS}}[X^2]$  and the right-hand one the young-cell limit  $\delta$ . Read the vertical extent of each curve as the size of the error made by treating  $p$  as a constant of the pathogen rather than of the epidemic's phase.

#### 1.5.4 Threshold and growth rate as functionals of one kernel

One object lies behind both (1.13) and the comparison of section 1.6. Define the cell-to-cell *generation kernel*

$$\mathcal{A}(\alpha) = \gamma T (g * e^{-c \cdot})(\alpha) = \gamma T \int_0^\alpha g(u) e^{-c(\alpha-u)} du, \quad (1.21)$$

the rate at which a cell of age  $\alpha$  generates new infected cells: it releases at rate  $g(u)$  at age  $u$ , a particle released then is still free a time  $\alpha - u$  later with probability  $e^{-c(\alpha-u)}$ , and infects at rate  $\gamma T$ . Its transform is  $\tilde{\mathcal{A}}(r) = \gamma T \tilde{g}(r)/(r + c)$ , so the characteristic equation (1.14) is exactly  $\tilde{\mathcal{A}}(r) = 1$ : the Euler–Lotka equation of a one-type age-structured population, with the infected cell as the individual and  $\mathcal{A}$  as its mean reproduction measure [17, 15].

Three things follow. First, the total mass of  $\mathcal{A}$  is the basic reproduction number,

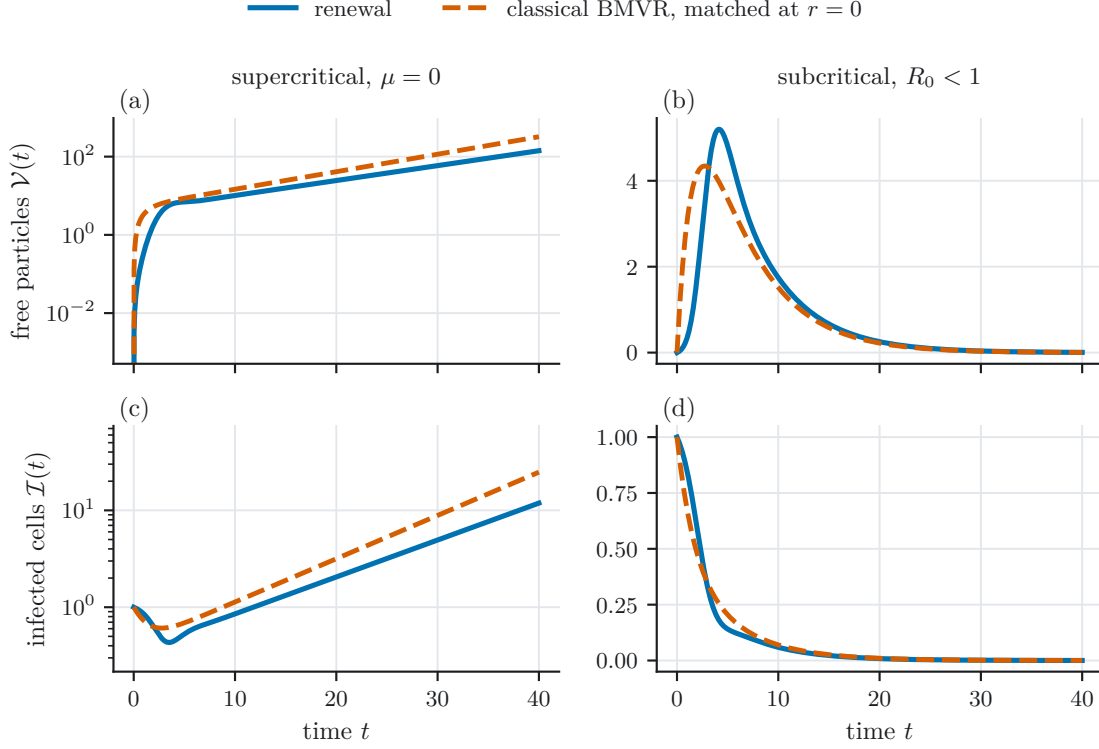
$$R_0 = \int_0^\infty \mathcal{A}(\alpha) d\alpha = \gamma T \int_0^\infty g \cdot \int_0^\infty e^{-cu} du = \frac{\gamma T V_\infty}{c}, \quad (1.22)$$

which depends on  $g$  only through its integral. Second, the normalised kernel  $\mathcal{A}/R_0$  is the *generation-time density*, the law of the age at which a cell produces a randomly chosen one of its offspring cells. Third, since  $r$  solves  $\tilde{\mathcal{A}}(r) = 1$  while  $R_0 = \tilde{\mathcal{A}}(0)$ , threshold and growth rate are different functionals of one kernel — the standard  $R_0$ -versus- $r$  distinction [18], here with both sides in closed form. The first two of these deliver the invariance that makes section 1.6 a fair comparison.

**Theorem 1.6 ( $R_0$  is unchanged by bursting).** *At fixed total per-cell output  $V_\infty$ , replacing constant budding by bursting leaves  $R_0$  exactly unchanged. Bursting alters the generation-time distribution — hence the growth rate  $r$ , hence the extinction probability, hence the stochastic early phase — but not the threshold itself.*

*Proof.* By (1.22),  $R_0 = \int_0^\infty \mathcal{A}$  depends on the release kernel only through  $\int_0^\infty g = V_\infty$ . Replacing bursting by budding at matched yield replaces  $g$  by another kernel with the same integral, and leaves  $R_0$  where it





**Figure 1.6:** Both state variables, both sides of the threshold. Free particles  $\mathcal{V}(t)$  (top row) and infected cells  $\mathcal{I}(t)$  (bottom) under the renewal system (solid) and under standard BMVR matched at  $r = 0$  (dashed), for a supercritical infection (left,  $R_0 = 1.76$ ) and a subcritical one (right,  $R_0 = 0.28$ ). The matched model gets the order of magnitude and not the shape; and the discrepancy is not confined to the release equation — age structure moves the cell trajectory too, which is why matching  $p$  alone cannot repair it. Parameters:  $(\lambda, \mu, \delta) = (1, 0, 0.1)$  throughout, with  $\gamma T = 0.04$ ,  $c = 0.25$  on the left and  $\gamma T = 0.01$ ,  $c = 0.4$  on the right.

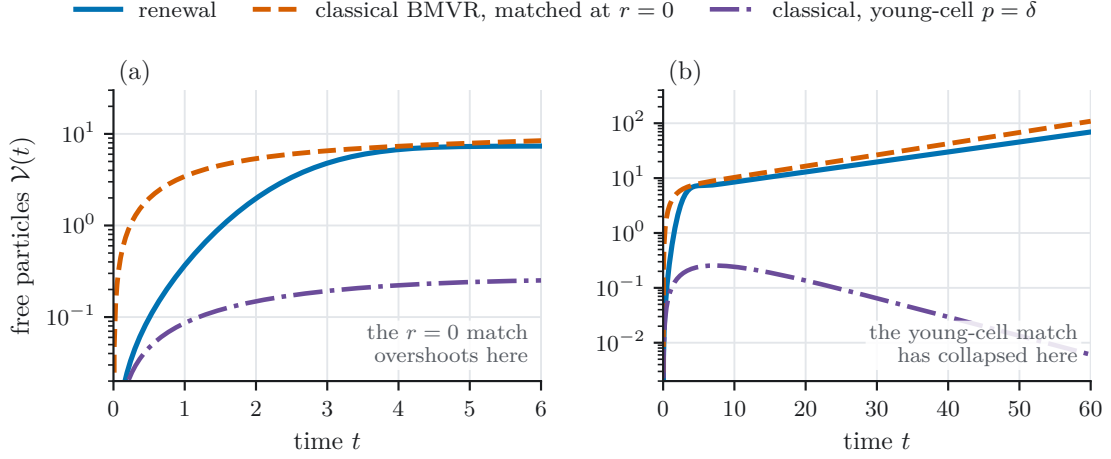
was. The generation-time density  $\mathcal{A}/R_0$ , by contrast, depends on the whole of  $g$ , and it is through that density that  $r$  is determined by  $\tilde{\mathcal{A}}(r) = 1$ .  $\square$

A caution attaches about assumption (A2): (1.22) is the *continuum* reproduction number, in which one virion contributes mean  $\gamma T/c$  new infections, whereas a discrete branching process absorbing virions on infection carries the mean  $m = qV_\infty$  with  $q = \gamma T/(\gamma T + c)$ . The two agree when  $\gamma T \ll c$ , and it is  $m$  that governs section 1.6.

### 1.5.5 What the projection loses

To see what age structure does to an epidemic, integrate both systems side by side. In figures 1.6 and 1.7 the solid curve is the renewal system and the dashed one standard BMVR with the fairest constant match, the  $r = 0$  end of (1.13), so that total lifetime output and mean productive lifetime agree.

The reading is consistent across panels. The matched model misses three things: the early lag, because young cells dominate and release is nearer the young-cell limit  $\delta$  than  $p_{\text{eff}}(0)$ ; the timing of the peak; and the late growth rate, since generation-time structure moves the operative release rate away from its  $r = 0$  value as the epidemic grows. The relative free-particle discrepancy is often order one or larger by the end of the plotted window in supercritical regimes, and worst when intracellular death is high, where  $V_\infty$  is small while the age structure of release stays strongly non-exponential.



**Figure 1.7:** Why no single constant pair will do. One renewal trajectory (solid) against two standard parameter choices: the  $r = 0$  match (dashed) and the young-cell limit  $p = \delta$ ,  $d_{\mathcal{I}} = \mu + \delta$  (dash-dot). (a) The first six time units, where the  $r = 0$  match overshoots by more than an order of magnitude while the young-cell parameters are the closer of the two. (b) The full window, where the ordering has reversed and the young-cell model has collapsed. Each choice is correct at one growth rate and the trajectory passes through many, exactly as (1.13) predicts. Both panels logarithmic in  $V$ . Parameters:  $(\lambda, \mu, \delta) = (1, 0, 0.1)$ ,  $\gamma T = 0.025$ ,  $c = 0.2$ ,  $R_0 = 1.38$ .

Note that the early lag is an eclipse phase that nobody put in. Within-host models routinely insert one by hand, as a fitted delay distribution or as explicit eclipse compartments [4, 5]; here it emerges from the intracellular dynamics with no extra parameter, its scale set by  $\mathbb{E}[T_{\text{prod}}]$ . The caution that goes with the observation is Herz et al.’s [5]: the shoulder in post-treatment viral load is a compound of pharmacological delay, intracellular delay, infected-cell decay and viral clearance, so the honest claim is not that this lag *is* the eclipse but that a lag of this shape arises without being fitted.

### 1.5.6 Identifiability: a negative result

How much of the intracellular process can a population-level fit recover? At any fixed growth rate  $r$ , the map

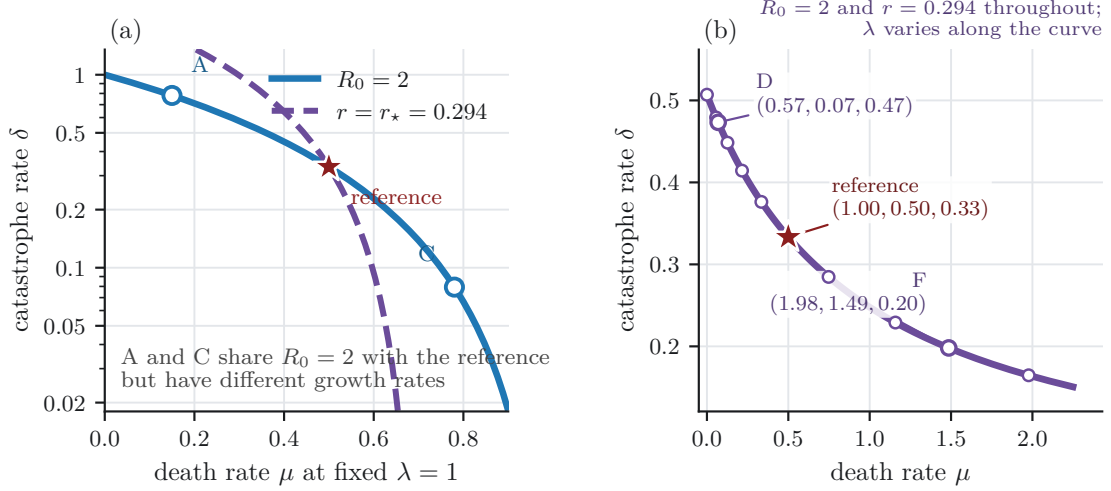
$$(\lambda, \mu, \delta) \mapsto (p_{\text{eff}}(r), d_{\mathcal{I}, \text{eff}}(r))$$

sends three rates to two numbers, and is therefore not invertible (figure 1.8). This is a structural identifiability statement of the kind Miao et al. [19] treat for viral dynamics generally, and it is negative for the same reason: the observable is a projection, and the projection has a kernel.

It is worth seeing that this result is corroborated from the fitting side. Mittler et al. [6] compared a range of intracellular delay distributions against HIV RNA data after protease inhibitor initiation and found the resulting parameter estimates largely insensitive to which distribution was assumed — a finding influential enough that the review of [4] cites it to justify preferring the simplest eclipse compartment. Read through theorem 1.3, that insensitivity is not a rival claim but the same one: on a single exponential phase a constant pair reproduces the kernel exactly, so the shape of the kernel is invisible in exactly the regime where those fits were performed. Consistently, when Kakizoe et al. [8] fitted in vitro SHIV data resolving the early transient, an Erlang delay did improve the fit.

A third observable is therefore required; in order of experimental accessibility:

1. the burst-size dispersion — the geometric ratio  $1/a$  fixes one combination of the rates directly;



**Figure 1.8:** Population observables leave level sets rather than identifying the intracellular rates. With  $c = 1$  and  $\gamma T = 1$ , the interior flooding point  $(\lambda, \mu, \delta) = (1, 0.5, 1/3)$  has  $R_0 = 2$  and  $r_* = 0.294$ . (a) One population observable leaves a level curve: in the slice  $\lambda = 1$  the contours  $R_0 = 2$  and  $r = r_*$  are distinct; alternatives A and C share  $R_0$  with the reference but not its growth rate. (b) With  $\lambda$  free, a continuous family of triples shares both. Joint knowledge of threshold and exponential growth therefore does not identify  $(\lambda, \mu, \delta)$ .

2. the fraction of cells that clear without bursting, which is  $b$ ;
3. the shape — not merely the mean — of the release-time density  $\delta J$ .

Any one of these, together with  $(p, d_T)$ , over-determines  $(\lambda, \mu, \delta)$  and converts the fit into a consistency test of the intracellular model. The limitation is not one of fitting procedure but of the information content of standard BMVR itself, and the remedy is the one [4] calls for in general terms: data at a finer scale of resolution, chosen to match the model's structure.

## 1.6 Bursting versus budding: the flooding parameter $L$

Does bursting help a pathogen get established? The question belongs to the stochastic early phase, where counts are small and extinction is a live possibility [20], and it is the question the stochastic intracellular model was built to answer: both versions of the early infection are branching processes whose offspring distributions are in closed form.

The mechanism at issue is worth naming before it is analysed. A bursting cell delivers its whole output in one instant, so a founder that survives to rupture presents the target population with many simultaneous infection attempts rather than a trickle spread over its lifetime. Whether that concentration helps or hurts is not obvious: many attempts at once buy insurance against the loss of any one particle, but they are all forfeited if the intracellular population dies out before the burst, which happens with probability  $b$ . We call the concentration *flooding*, and the question is when it pays.

### 1.6.1 The offspring of one infected cell

In the branching approximation each released particle independently either infects a target cell, at rate  $\gamma T$ , or is cleared, at rate  $c$ , and so infects with probability

$$q = \frac{\gamma T}{\gamma T + c}. \quad (1.23)$$

The offspring of one infected cell is the burst size  $\mathcal{K}$  thinned by  $q$ . Summing (1.46), the unconditional burst-size probability generating function is  $G_{\mathcal{K}}(z) = b + (\delta/\lambda)z/(a - z)$  — including the mass  $b$  of cells that never burst — and thinning replaces  $z$  by  $y := 1 - q + qz$ :

$$G_{\text{off}}(z) = b + \frac{\delta}{\lambda} \frac{y}{a - y}, \quad y = 1 - q + qz. \quad (1.24)$$

One checks  $G_{\text{off}}(1) = b + (\delta/\lambda)/(a - 1) = b + B = 1$ . The mean and variance of the offspring number are

$$m = G'_{\text{off}}(1) = qV_{\infty}, \quad \text{Var}_{\text{burst}} = \frac{2q^2V_{\infty}}{a - 1} + qV_{\infty} - q^2V_{\infty}^2. \quad (1.25)$$

When  $\gamma T \ll c$ ,  $q \approx \gamma T/c$  and  $m \approx R_0$ .

The comparator is the budding cell of section 1.2.2, whose output is geometric with mean  $p/d_{\mathcal{I}}$ . Match it to the bursting cell's total output,  $p/d_{\mathcal{I}} = V_{\infty}$ . That matching is what makes the comparison fair, and by theorem 1.6 it costs nothing: at equal total output the two mechanisms have exactly the same  $R_0$ , so everything that follows compares two pathogens at one threshold, and any difference between them is a difference in how the output is *scheduled* rather than in how much of it there is. Thinning a geometric law leaves a geometric law, so the budding offspring number is geometric with the same mean  $m = qV_{\infty}$ .

### 1.6.2 Extinction, and the flooding advantage

The extinction probability of a branching process is the smallest root of  $G_{\text{off}}(z) = z$  in  $[0, 1]$ . With  $y = 1 - q + qz$  the fixed-point equation is quadratic,  $z = 1$  is always a root, and the other root gives, for  $m > 1$ ,

$$z_{\text{ext}}^{\text{burst}} = \frac{a(1 - q + qb) - 1 + q}{q}, \quad z_{\text{ext}}^{\text{bud}} = \frac{1}{m}. \quad (1.26)$$

Everything now turns on a single combination of the intracellular rates.

**Definition 1.7** (The flooding parameter).

$$L := a(1 - b) = a - \frac{\mu}{\lambda}, \quad \text{so that} \quad V_{\infty} = \frac{L}{a - 1}. \quad (1.27)$$

**Theorem 1.8** (The flooding advantage). *For  $m > 1$ ,*

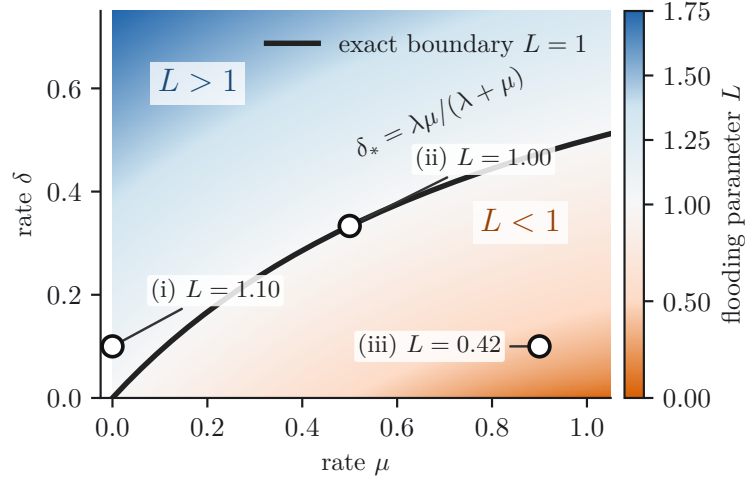
$$z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L - 1) \left( \frac{1}{m} - 1 \right). \quad (1.28)$$

*Hence bursting strictly lowers the establishment extinction probability, relative to matched-mean budding, if and only if  $L > 1$ , that is*

$$\delta(\lambda + \mu) > \lambda\mu \quad \Longleftrightarrow \quad \frac{1}{\delta} < \frac{1}{\lambda} + \frac{1}{\mu}, \quad (1.29)$$

*with the convention that the condition holds for all  $\delta > 0$  when  $\mu = 0$ . When  $L = 1$  the two extinction probabilities are identical for every  $q$ .*

*Proof.* The identity (1.28) is algebra: substitute (1.26),  $m = qV_{\infty} = qL/(a - 1)$  and simplify. Since  $m > 1$  makes the second factor negative, the sign of the difference is the sign of  $1 - L$ . For the criterion,  $L > 1$



**Figure 1.9:** Where a pathogen sits relative to the flooding boundary. The flooding parameter  $L(\mu, \delta)$  at  $\lambda = 1$ ; the exact curve  $L = 1$ , equivalently  $\delta_* = \lambda\mu/(\lambda + \mu)$ , partitions the rate plane into a region where bursting lowers extinction risk and one where it raises it. The three marked points are the regimes of [table 1.2](#),  $(\mu, \delta) = (0, 0.1)$ ,  $(0.5, 1/3)$  and  $(0.9, 0.1)$ , with  $L = 1.10$ ,  $1.00$  and  $0.42$ .

is  $a(1 - b) > 1$ , i.e.  $a > 1 + \mu/\lambda$ ; inserting the definition of  $a$  from (1.5) and squaring — legitimate since both sides are positive, or the right-hand side is not — yields  $(\lambda + \mu + \delta)^2 - 4\lambda\mu > 4\lambda^2(1 + \mu/\lambda - \eta)^2$ , which reduces to  $\delta(\lambda + \mu) > \lambda\mu$ .  $\square$

The criterion (1.29) is the chapter’s central biological statement, and it reads cleanly as a comparison of timescales. The burst clock  $1/\delta$  must be shorter than the harmonic-type combination  $(1/\lambda + 1/\mu)$  of the birth and death timescales. A cell that takes too long to rupture is a cell whose internal population has time to die out first, and the all-or-nothing hazard  $b$  then outweighs the insurance that a simultaneous batch buys. When  $\mu = 0$  that hazard vanishes and bursting wins at every  $\delta$ . [Figure 1.9](#) shows where the boundary lies in the rate plane; [table 1.5](#) confirms the identity in each regime, and [figure 1.10](#) traces both extinction probabilities across the whole range of  $q$ .

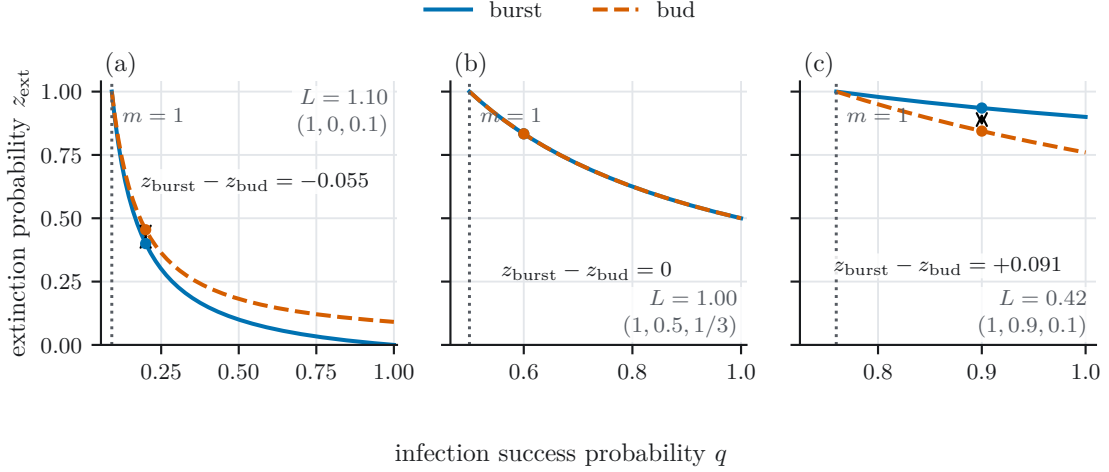
| $(\lambda, \mu, \delta)$ | $q$ | $z_{\text{ext}}^{\text{burst}}$ | $z_{\text{ext}}^{\text{bud}}$ | verdict                      |
|--------------------------|-----|---------------------------------|-------------------------------|------------------------------|
| $(1, 0, 0.1)$            | 0.2 | 0.400                           | 0.455                         | bursting wins ( $L = 1.10$ ) |
| $(1, 0.9, 0.1)$          | 0.9 | 0.935                           | 0.844                         | budding wins ( $L = 0.42$ )  |
| $(1, 0.5, 1/3)$          | 0.6 | 0.833                           | 0.833                         | tie ( $L = 1$ )              |
| $(1, 0.5, 1/3)$          | 0.9 | 0.556                           | 0.556                         | tie ( $L = 1$ )              |

**Table 1.5:** Extinction probabilities from (1.26) and the sign of their difference. The gap is  $(L - 1)(1/m - 1)$  in every row.

For  $m \leq 1$  both processes die out almost surely and the comparison is empty; (1.28) changes sign there, an artefact of both formulae running outside their domain  $[0, 1)$ , not a reversal of the biology.

### 1.6.3 The boundary $L = 1$ : a coincidence of laws

The boundary case is more than a tie of extinction probabilities.



**Figure 1.10:** Extinction probabilities for bursting and matched-mean budding across the three regimes of [table 1.2](#). Each pair of curves begins at  $m = 1$ ; the signed gap follows [\(1.28\)](#). The  $q$ -windows differ because  $m = 1$  sits at different  $q$  in each regime.

**Proposition 1.9.** *The following are equivalent:*

$$L = 1 \iff a = 1 + \frac{\mu}{\lambda} \iff \delta(\lambda + \mu) = \lambda\mu \iff \frac{1}{\delta} = \frac{1}{\lambda} + \frac{1}{\mu}. \quad (1.30)$$

When they hold, the bursting offspring law [\(1.24\)](#) is exactly geometric with mean  $m$  — not merely equal in extinction probability to the budding comparator. In particular the offspring variances coincide, and in general

$$\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}} = \frac{2q^2 V_{\infty}(L - 1)}{a - 1}, \quad (1.31)$$

so that at matched mean the bursting offspring variance is smaller than the budding one when  $L > 1$ , equal when  $L = 1$ , and larger when  $L < 1$ .

*Proof.*  $L = 1$  is  $a(1 - b) = 1$ , i.e.  $a - 1 = ab = \mu/\lambda$ , giving the chain of equivalences. Under it  $b = \delta/\lambda$ , and [\(1.24\)](#) collapses:

$$G_{\text{off}}(z) = b \left( 1 + \frac{y}{a - y} \right) = \frac{ab}{a - y} = \frac{a - 1}{a - y} = \frac{(a - 1)/a}{1 - y/a},$$

using  $ab = a - 1$ . With  $y = 1 - q + qz$ ,  $1 - y/a = \frac{a-1+q}{a} \left( 1 - \frac{q}{a-1+q}z \right)$ , so

$$G_{\text{off}}(z) = \frac{1 - \varrho}{1 - \varrho z}, \quad \varrho = \frac{q}{a - 1 + q} = \frac{m}{1 + m}, \quad m = \frac{q}{a - 1} = qV_{\infty},$$

the probability generating function of a geometric law on  $\{0, 1, 2, \dots\}$  with mean  $m$  — the same law as the matched budding comparator. The variance identity [\(1.31\)](#) is the difference between the geometric variance  $m(1 + m)$  and [\(1.25\)](#), with  $V_{\infty} = L/(a - 1)$ .  $\square$

So  $L$  does more than fix a sign. At the boundary, bursting and budding are the same branching process in different clothes; away from it, [\(1.31\)](#) says precisely how the two offspring laws differ, and the extinction ordering of [theorem 1.8](#) is the ordering of their variances at fixed mean. That is the mechanism behind the criterion: concentrating output into one batch reduces offspring variance — which helps at establishment — but only if the batch is delivered often enough to offset the mass  $b$  at zero.

### 1.6.4 The price: invasion speed

Establishment is half the story; the other half is how fast an established infection grows. Match the models a second time, at  $R_0$  and at mean productive lifetime: take the budding cell with  $d_{\mathcal{I}} = 1/\mathbb{E}[T_{\text{prod}}]$  and  $p = V_{\infty}d_{\mathcal{I}}$  — exactly  $p_{\text{eff}}(0)$ , the fairest constant match available — and choose  $\gamma T$ , or  $c$ , so that both share  $R_0 = 2$ . Solving for the growth rates,  $r_{\text{bud}}$  from  $(r + d_{\mathcal{I}})(r + c) = \gamma T p$  and  $r_{\text{burst}}$  from (1.14), gives the same ordering in all three regimes:

$$r_{\text{bud}} > r_{\text{burst}}. \quad (1.32)$$

| $(\lambda, \mu, \delta)$ | $L$  | $d_{\mathcal{I}}$ | $r_{\text{bud}}$ | $r_{\text{burst}}$ |
|--------------------------|------|-------------------|------------------|--------------------|
| (1, 0, 0.1)              | 1.10 | 0.417             | 0.250            | 0.180              |
| (1, 0.5, 1/3)            | 1.00 | 0.910             | 0.395            | 0.294              |
| (1, 0.9, 0.1)            | 0.42 | 0.701             | 0.343            | 0.198              |

**Table 1.6:** Deterministic growth rates at matched  $R_0 = 2$ , matched total yield and matched mean productive lifetime, with  $c = 1$ . Budding invades faster on both sides of the flooding boundary and on it, so the ordering is not the same effect as [theorem 1.8](#).

Whether (1.32) holds in general is not settled here, but it reduces to a single inequality in which the coupling constants do not appear.

**Proposition 1.10** (The ordering is a Laplace-transform comparison). *Match the comparator as above, so that  $\int_0^{\infty} g_{\text{bud}} = \int_0^{\infty} g_{\text{burst}} = V_{\infty}$ . Then  $r_{\text{bud}} > r_{\text{burst}}$  for every pair  $(\gamma T, c)$  with  $R_0 > 1$  if and only if*

$$\tilde{g}_{\text{burst}}(r) \leq \tilde{g}_{\text{bud}}(r) = \frac{p}{r + d_{\mathcal{I}}} \quad \text{for all } r > 0, \quad (1.33)$$

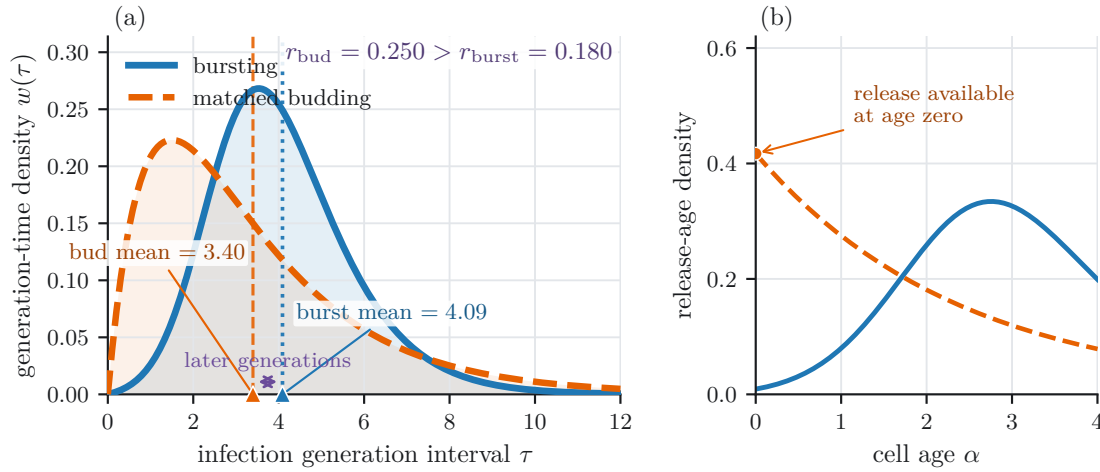
*that is, if and only if the release age under bursting dominates the matched exponential release age in the Laplace-transform order.*

*Proof.* Each model's growth rate is the unique root of  $\phi(r) := \gamma T \tilde{g}(r) - (r + c)$ ;  $\phi$  is strictly decreasing, and  $\phi(0) = \gamma T V_{\infty} - c = c(R_0 - 1) > 0$ , so the root is positive and unique. If (1.33) holds then  $\phi_{\text{burst}} \leq \phi_{\text{bud}}$  pointwise, so the root of the former is no larger. Conversely, suppose  $\tilde{g}_{\text{burst}}(r_0) > \tilde{g}_{\text{bud}}(r_0)$  for some  $r_0 > 0$ . Choose any  $c > 0$  and set  $\gamma T = (r_0 + c)/\tilde{g}_{\text{burst}}(r_0)$ , so that  $r_{\text{burst}} = r_0$ ; then  $\phi_{\text{bud}}(r_0) < 0$  and  $r_{\text{bud}} < r_0$ . Taking  $c$  small makes  $R_0 = \gamma T V_{\infty}/c$  exceed one, so the ordering fails for an admissible coupling.  $\square$

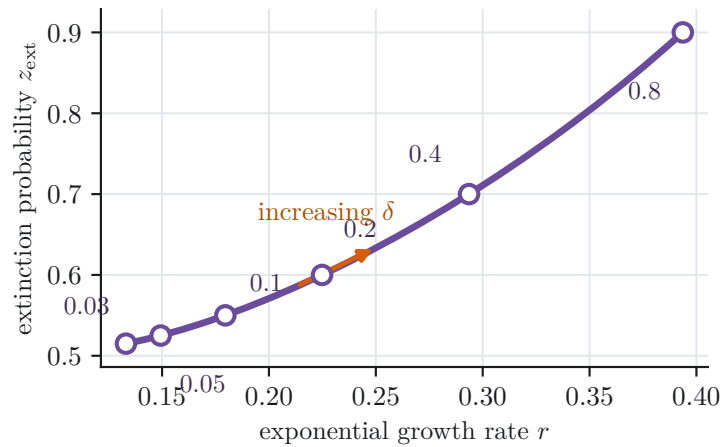
Both sides of (1.33) agree at  $r = 0$ , by the matching, so its content is entirely in how the two release schedules are spread over age. The inequality was checked at 1500 triples over  $(\mu, \delta) \in [10^{-3}, 10] \times [10^{-3}, 16]$  at  $\lambda = 1$ , at 60 values of  $r$  across four decades, without a violation (test N of the verification suite). The *usual* stochastic order does not hold — the burst release-age survival function dips below the matched exponential — so (1.33) is not a weaker way of saying something stronger. It is the order the problem actually has.

The mechanism is generation time in the exact sense of [section 1.5.4](#). The budding cell starts releasing at once while the bursting cell must first accumulate load, so budding's generation-time density  $\mathcal{A}/R_0$  sits earlier, and by  $\tilde{\mathcal{A}}(r) = 1$  an earlier density gives a larger root. At  $(\lambda, \mu, \delta) = (1, 0, 0.1)$  with  $c = 1$  the mean generation time is  $2.398 + 1 = 3.398$  for budding against  $3.089 + 1 = 4.089$  for bursting, which is what puts  $r_{\text{bud}} = 0.250$  above  $r_{\text{burst}} = 0.180$  ([figure 1.11](#)). This is the same delay that appeared as the eclipse-like lag of [section 1.5.5](#), and it is the phage lysis-timing problem in another notation: holding on longer makes more particles per cell but takes longer to make them [21, 18]. That an infectious-period distribution shifts the growth rate at fixed threshold is a general fact about renewal epidemics rather than anything peculiar to bursting [22, 23]; what is particular here is that the distribution is derived rather than assumed.





**Figure 1.11:** The generation-time structure behind the first row of [table 1.6](#). At  $(\lambda, \mu, \delta) = (1, 0, 0.1)$ , budding and bursting are matched at  $R_0 = 2$ ,  $c = 1$ , total yield and mean productive lifetime. The main curves are the normalised infection-generation laws  $\mathcal{A}(\alpha)/R_0$ , including the common extracellular waiting-time component; panel (b) isolates release age and shows why the budding law leads.



**Figure 1.12:** Tuning  $\delta$  traces a strict extinction–growth trade-off, drawn parametrically as  $\delta$  varies from 0.03 to 0.8 with  $\lambda = 1$ ,  $\mu = 0$ ,  $c = 1$  and  $R_0 = 2$ . The coupling is reset along the curve to hold  $R_0 = 2$ , and the budding comparator is matched in lifetime yield  $V_\infty$  and in mean productive lifetime. There is no setting of  $\delta$  that improves both coordinates: faster growth is bought with a higher extinction probability.

### 1.6.5 The trade-off

Put together, the two comparisons say what bursting buys and what it costs. At fixed total output per cell, bursting leaves  $R_0$  untouched, lowers the probability of early extinction when  $L > 1$ , and slows deterministic invasion. A pathogen tuning  $\delta$  therefore faces a genuine trade-off ([figure 1.12](#)):  $z_{\text{ext}}$  and  $r$  move in opposite directions, and no setting of  $\delta$  improves both. For  $\mu > 0$ , budding beats bursting on extinction until  $\delta$  crosses  $\delta_* = \lambda\mu/(\lambda + \mu)$ , beyond which bursting takes over; for  $\mu = 0$  bursting wins at every  $\delta$ , at the continuing cost of a slower  $r$ .

Two remarks place the result. First, its scope: both comparisons hold the offspring *mean* fixed and are branching-process statements, so they say nothing about nonlinear effects, where other advantages of flooding may live ([section 1.8.2](#)). Second, its company. Lord and Bonsall [\[24\]](#) reach a qualitatively parallel conclusion for within-host virus evolution by a different route, using fixed-delay models to compare virion release strategies and finding that lysis-like strategies are favoured when the terminal event comes quickly and budding when it is slow. The criterion



(1.29) is a sharp version of “quickly”, written in intracellular rates, and it comes with the exact size of the advantage from (1.28). The evolutionary framing belongs to the chapter on bursting and budding pathogenesis, which inherits from this comparison the awkward fact that one mechanism can be favoured on establishment and disfavoured on invasion speed at once — the nested-model problem in its usual form [25].

## 1.7 The reach of the construction

Section 1.4.3 showed that any pair of kernels induces the same population equations. Bursting and constant budding are the ends of a spectrum; between them lie mechanisms that release many times per cell or release part of the load, and beyond it lie models that are not birth–death–catastrophe processes at all. All supply a pair  $(S, g)$ , and none changes (1.11). This section works through the two intermediates that matter and then says where the generality stops.

### 1.7.1 Recurrent release: the reset process

The absorbing process releases at most once per cell. The intermediate worth isolating lets a cell release *many* times: while the cell is alive the load undergoes birth  $\lambda n$ , death  $\mu n$ , optional immigration  $\nu$ , and a *dump that resets* — at rate  $\delta n$  the whole pool is released and the load returns to zero, after which it may re-accumulate — with the cell dying at a separate, load-independent rate  $d_I$ . Reading the load as a local assembled pool rather than the whole-cell contents makes this a tractable multi-pulse caricature between full dumping and continuous export.

Because dumps do not absorb, the kernels change but the skeleton does not:

$$S(\alpha) = e^{-d_I \alpha}, \quad g(\alpha) = \delta e^{-d_I \alpha} \mathbb{E}[X_\alpha^2], \quad (1.34)$$

with  $\mathbb{E}[X_\alpha^2]$  computed for the non-absorbing reset process. Linear rates keep its moments tractable, so  $g$  is explicit once they are solved, and structurally  $g \propto \mathbb{E}[X^2]$  exactly as for the absorbing process. The effective parameters become

$$p_{\text{eff}}(r) = (r + d_I) \tilde{g}(r), \quad d_{I, \text{eff}} \equiv d_I, \quad (1.35)$$

and  $R_0 = (\gamma T/c) \int_0^\infty g$ , finite only for  $d_I > 0$  since otherwise recurrent dumps could continue forever. If the reset process admits a stationary law  $\pi$  and  $d_I$  is small, the mature-cell limit is

$$p_\infty = \lim_{\alpha \rightarrow \infty} \frac{g(\alpha)}{S(\alpha)} = \delta \mathbb{E}_\pi[X^2], \quad (1.36)$$

the same form as the old-cell limit (1.17) of the absorbing process. A long-lived infected cell therefore behaves as constant- $p$  BMVR with that  $p$ , while transients — first dumps, young cells — still need the full kernel. This is what makes the spectrum claim concrete rather than gestural: an intermediate with kernels, an effective release rate, a mature limit, and a definite answer as to which standard model it converges on.

### 1.7.2 Partial release, and antiviral mode of action

The remaining axis is *how much* leaves the cell when a release event fires. The natural intermediate removes a random part of the load; the default is boolean thinning, each unit released independently with probability  $\varphi$ . At the mean-field level with event clock  $\delta X$ , writing  $Q$  for the total intracellular stock across productive cells, the population equations are

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_I \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \delta \varphi \frac{Q^2}{\mathcal{I}} - d_I Q, \quad \frac{d\mathcal{V}}{dt} = \delta \varphi \frac{Q^2}{\mathcal{I}} - c \mathcal{V}. \quad (1.37)$$

With a load-independent clock  $\lambda(X) = \delta$  instead, the mean export is  $\delta\varphi Q$  and the equations collapse to stock-and-export dynamics with  $\varepsilon_{\text{eff}} = \delta\varphi$ . So partial release is invisible in the mean-field equations whenever the event clock is load-independent, and visible when it is load-dependent; in single-cell release-count distributions it is always visible.

This axis is where the construction meets the second of the two payoffs that [4] identifies for intracellular modelling: the ability to locate a drug’s mode of action at a named step of the viral life cycle. In the multiscale hepatitis C models [9, 10] a direct-acting antiviral is represented by three separate factors — blocking intracellular RNA production by  $(1 - \epsilon_a)$ , blocking packaging and secretion by  $(1 - \epsilon_s)$ , and enhancing intracellular degradation by a factor at least one. Each has a counterpart here. Blocking production lowers  $\lambda$ ; enhancing degradation raises  $\mu$ ; blocking entry lowers  $\gamma T$ ; and blocking secretion is exactly a reduction in  $\varphi$ , since  $\varphi = 1 - \epsilon_s$ . Every one of these is a transformation of  $(S, g)$ , so a drug in this framework does not add parameters to the population layer but moves the kernels, and its population signature is whatever (1.13) makes of the new pair.

Two consequences follow, both in principle testable. The first is that a drug moves the pathogen relative to the flooding boundary, and the direction is read off (1.29) without further calculation. Enhancing intracellular degradation raises  $\mu$  and so lowers  $1/\lambda + 1/\mu$ , moving the pathogen toward the  $L < 1$  side on which its burst schedule is a liability rather than an asset at establishment; blocking intracellular production raises  $1/\lambda$  and moves it the other way. Neither move need cross the boundary — (1.29) is an inequality, not a monotone score — but the sign of the shift is fixed, and (1.28) gives the size of what is gained or lost at each end. The second is that because  $p_{\text{eff}}$  depends on  $r$ , an efficacy inferred from a decay slope inherits the same phase dependence that section 1.5.3 finds in  $p$  itself, which is one reading of why the hepatitis C clearance estimates of section 1.8.1 disagreed across therapies. Whether the flooding advantage survives at intermediate  $\varphi$ , and whether the closed forms persist there at all, is open.

### 1.7.3 Beyond birth–death–catastrophe: HIV

The results above are written for pathogens whose intracellular life ends in load-proportional catastrophe. HIV is not one: it buds from the plasma membrane, and an infected cell need not rupture in a single event. After integration the template count is one or a few proviruses and structural products do not self-replicate, so the mean load follows an immigration-like  $\bar{x}' \approx \nu - \mu\bar{x}$  rather than the multiplicative growth that produces  $\delta K$ . Yet limiting-dilution data show something surge-like — long quiet periods, then a concentrated bout of release — which is the phenomenon the catastrophe process exists to explain. Hataye et al. [26] report first detection around day 3 with bulk production over days 3–5, an Erlang eclipse with  $n \approx 5$  stages, many silent lineages, and a sigmoid rather than linear dependence of establishment on the number of virus-releasing cells, with a threshold near 2.3 cells.

The position taken here is neither to force HIV into the catastrophe process nor to retreat to constant- $p$  BMVR. The skeleton (1.11) carries HIV on a different pair of kernels:  $S_{\text{HIV}}$  the probability of still being in a productive lineage and  $g_{\text{HIV}}$  the density of productive sojourn times, with a multi-stage eclipse supplying the delay — near zero for small  $\alpha$ , then rising and falling. The effective parameters carry over verbatim under  $(I_{\text{fix}}, g) \mapsto (S_{\text{HIV}}, g_{\text{HIV}})$ . What does not carry over is independent branching near the observed threshold: the data favour a sigmoid incidence such as

$$f(\mathcal{V}, T) = \gamma T \mathcal{V} \cdot \frac{\mathcal{V}}{K_A + \mathcal{V}}, \quad (1.38)$$

inefficient at very low  $\mathcal{V}$  and saturating to mass action for  $\mathcal{V} \gg K_A$ . It is worth noting that this is the opposite curvature at low  $\mathcal{V}$  from the saturating alternatives catalogued in [4] — Holling type 2, Beddington–DeAngelis, Crowley–Martin — which depress infection at *high* viral load; the departure from mass action wanted here is an Allee effect at establishment, not saturation at

saturation. Under (1.38),  $R_0$  remains the large-population threshold while early establishment is governed in addition by the scale  $K_A$ , and [proposition 1.2](#) no longer holds exactly, since a nonlinear incidence breaks the moment closure.

#### 1.7.4 Where the reach stops

What carries over to any single-cell mechanism is (1.11), the map  $p_{\text{eff}} = \tilde{g}/\tilde{S}$ , the reading of constant  $p$  as an exponential-phase projection, and the generation-kernel formulation of  $R_0$  and  $r$ . What does not is everything resting on the closed forms of the birth–death–catastrophe process: the flooding criterion  $1/\delta < 1/\lambda + 1/\mu$ , the geometric burst law and its identification with the quasi-stationary distribution, and one-shot cumulative release as the only single-cell trajectory. For HIV in particular, independent branching is not the full establishment story near the observed threshold; the flooding comparison survives as a *linear* baseline, at matched mean yield with offspring variances compared, but synergy can dominate variance effects there and the two mechanisms must be separated before either is invoked.

## 1.8 Discussion

This chapter took a completed stochastic single-cell theory and embedded it in a population. What came out is a renewal model of within-host infection forced by the intracellular process and exact in its kernels; the demonstration that the standard model is a projection of it, valid one exponential phase at a time; an effective-parameter map whose two ends differ by the factor  $\mathbb{E}[X^2]_{\text{QS}}$ ; and a closed-form answer to whether bursting pays, in which the flooding parameter  $L = a(1 - b)$  decides the sign and  $(L - 1)(1/m - 1)$  gives the size.

### 1.8.1 Consequences for fitting practice

Three results bear on how within-host models are fitted, and each says what to do rather than only what is true.

**A fitted  $p$  is a property of the phase.** Acute-phase and set-point estimates should therefore differ systematically, in a direction and by an amount (1.13) specifies. Two published estimates of  $p$  for the same organism at different phases are not in conflict, and the ratio between them is itself an observable: by (1.20) it cannot exceed  $\mathbb{E}[X^2]_{\text{QS}}$ , so a reported spread larger than that is evidence against the intracellular model rather than against either fit. There is precedent for the general shape of this argument. Hepatitis C clearance was estimated at  $c = 6.2/\text{day}$  from interferon- $\alpha$  trials [27] and at  $c = 23.3/\text{day}$  from direct-acting antiviral data [9] — close to a fourfold spread in a rate that cannot depend on the therapy used to measure it. The resolution was not to adjudicate between the fits but to add intracellular structure, after which the two biphasic declines turn out to be reporting different pairs of underlying rates: under interferon- $\alpha$  the slopes are the intracellular loss rate and the infected-cell death rate, and neither is  $c$ , whereas under direct-acting antivirals the fast slope is  $c$  after all [10]. The earlier figure was not a bad measurement of clearance; it was a good measurement of something else. That is the same error [section 1.5](#) identifies for  $p$ , and it is the general lesson: a slope read off a transient is not a rate of the organism until a model says which rate it is.

**The eclipse may be being fitted twice.** The lag that within-host models insert by hand emerges here from the intracellular dynamics with no extra parameter, its scale set by  $\mathbb{E}[T_{\text{prod}}]$ . A model already carrying a fitted eclipse and a bursting pathogen is therefore fitting the same delay twice, and the two estimates should be checked against each other before either is trusted.

**A population fit needs a single-cell companion.** No fit of  $(p, d_I)$  recovers the three intracellular rates (section 1.5.6), so a population fit is worth doing only alongside one of the three single-cell observables named there. With any one of them the system is over-determined and the fit becomes a test of the intracellular model rather than a description of it. This is a concrete instance of the general call in [4] for data collected at a finer scale of resolution to validate complex models: the chapter names which three measurements would do it, and in what order of experimental accessibility.

### 1.8.2 Limitations

The assumptions of section 1.1.1 are load-bearing. Holding target cells constant (A1) makes the model linear, so supercritical solutions grow without bound and peak times exist only in subcritical cases; depletion would cap them, but the linear analysis remains the right object for establishment and early growth, where the chapter's claims live. The continuum convention (A2) is why  $R_0 = \gamma TV_\infty/c$  sits alongside the branching mean  $m = qV_\infty$ : they agree when  $\gamma T \ll c$  and are otherwise different quantities, and confusing them would misread theorem 1.8. The synchronous initial condition (A3) is not innocuous, since a non-trivial initial age distribution adds a term to (1.9).

Single infection (A4) is the most substantive restriction. Multiply infected cells release superlinearly more — (1.49) gives  $V_\infty^{(k)}$  — and multiplicity is not rare: as many as a tenth of HIV-infected cells are thought to be multiply infected, and coinfection shapes estimates of infected-cell death and production, drug-resistance evolution and immune escape [4]. The standard remedy adds compartments for doubly and triply infected cells, which the same review notes is difficult to validate against plasma titres because of the parameter count. The MOI- $k$  row of table 1.4 suggests a cheaper route: multiplicity enters through the kernels  $(I^k - D^k, \delta K_k)$  while the skeleton stays one-dimensional, adding no compartments and no fitted parameters. Carrying those kernels through the population layer has not been done here, and it is the most obvious extension. It would also connect to the alternative low-dimensional treatment of coinfection due to Koelle et al. [28], whose macroparasite-style bookkeeping tracks the total internalised virus across all cells — the quantity written  $Q$  in (1.37) — and which likewise abandons the standard assumption that burst size is independent of the extent of coinfection. Absence of spatial structure (A5) is the most consequential omission for a pathogen that spreads cell to cell.

Finally, the chapter is written in dimensionless rates with  $\lambda = 1$  and does not say where a real pathogen sits on figure 1.9. For *Y. pestis* in the macrophage the intracellular interval is short and the biology resists a clean measurement: the bacterium enters a permissive vacuole and replicates there while the temperature shift from the flea to the host induces the capsule, the pH 6 antigen and the type III secretion system, and then leaves by a route the primary literature still treats as unresolved, with apoptosis, pyroptosis and necroptosis all reported. Placing that organism relative to  $L = 1$  needs  $\lambda$ ,  $\mu$  and  $\delta$  for the intracellular phase — and the exit mechanism is exactly what fixes  $\delta$ .

### 1.8.3 Open problems

1. **Two-type killed second moments.** The release kernel of a two-type renewal model needs

$$\mathbb{E}[X_\alpha^2 \mathbf{1}\{\tau_c > \alpha\}], \quad \mathbb{E}[X_\alpha Y_\alpha \mathbf{1}\{\tau_c > \alpha\}], \quad \mathbb{E}[Y_\alpha^2 \mathbf{1}\{\tau_c > \alpha\}],$$

together with the two-type analogue of  $D$ . The moment hierarchy does not close upward, exactly as in one type; the one-type escape was to differentiate the Riccati equation, and the two-type route should be to differentiate the backward system of the two-type chapter.

This is the main technical obstacle between this chapter and an eclipse-aware renewal model, and it should not be assumed easy.

2. **The flooding boundary against real parameters.** Locate *Y. pestis* in the macrophage relative to  $\delta(\lambda + \mu) > \lambda\mu$  and compute the size of the advantage from (1.28). The estimates discussed in the chapter on its distribution theory — the near-certain bursting of *F. tularensis* [29], the clearance-dominated anthrax case — suggest real pathogens sit on both sides of the boundary, which is what makes the question worth asking.
3. **Population-level variance.** Propagate  $\text{Var}(W)$  through the branching structure to a whole-host prediction. For single-cell assays the first moments suffice; for whole-host data the variance is the main stochastic signal that standard BMVR cannot reproduce, and the integer-valued nature of the offspring count enters through the same quantity [30].
4. **Multiplicity through the population layer, and partial release:** where in  $\varphi$  the flooding advantage survives.

Two smaller questions are single definite statements rather than open directions. The hypothesis of proposition 1.5 — that  $\rho = g/S$  increases with cell age — is a claim about the one closed-form function (1.19) of one variable, so proving it is finite algebra; and the ordering  $r_{\text{bud}} > r_{\text{burst}}$  is by proposition 1.10 equivalent to the single Laplace inequality (1.33), with the coupling constants eliminated. Both are supported by wide numerical sweeps and neither is proved.

#### 1.8.4 Connections forward

The two-type process of the two-type chapter supplies the most immediate continuation, and it is where the principal technical obstacle lives. In the regime  $\delta_1 = 0 < \delta_2$  — no burst before conversion — the no-catastrophe probability has zero initial hazard and then rises: precisely the eclipse-phase signature that within-host models insert by hand, emerging mechanistically, with the eclipse duration set by the conversion rate. Nothing in (1.11) need change to accept those kernels when they exist. The bursting–budding comparison finds its evolutionary home in the chapter on bursting and budding pathogenesis, where the intermediate mechanisms of section 1.7 supply the phenotypes a selection argument needs. The nonlinear effects that the later modelling chapters treat — immune saturation, local depletion, the all-run-out-at-once intuition — are exactly what the branching comparison deliberately excludes, and should be named wherever the flooding advantage is invoked.

## 1.A Single-cell results quoted from earlier chapters

Write  $w = e^{\theta t}$ . The roots of  $\lambda f^2 - (\lambda + \mu + \delta)f + \mu$  are  $a, b$  of (1.5), with  $b < 1 < a$ , and the derived constants are

$$A = a - 1, \quad B = 1 - b, \quad \theta = \lambda(a - b), \quad \kappa = 1 + \frac{\delta}{2\lambda}. \quad (1.39)$$

The identities that simplify everything downstream are

$$ab = \frac{\mu}{\lambda}, \quad a + b = \frac{\lambda + \mu + \delta}{\lambda}, \quad AB = \frac{\delta}{\lambda}, \quad A + B = a - b. \quad (1.40)$$

The no-burst probability, the internal-extinction probability and the non-fixation probability are

$$\begin{aligned} I(t) &= \frac{aB + bAw}{B + Aw}, & D(t) &= \frac{ab(w - 1)}{aw - b}, \\ I_{\text{fix}}(t) &= I(t) - D(t) = \frac{(a - b)^2 w}{(B + Aw)(aw - b)}. \end{aligned} \quad (1.41)$$

Here  $I(\infty) = D(\infty) = b$ , the probability that the cell clears internally without ever bursting, while  $I_{\text{fix}}(0) = 1$ ,  $I'_{\text{fix}}(0) = -(\mu + \delta)$  and  $I_{\text{fix}}(\infty) = 0$ . With  $I' = -\delta J$  and  $V' = \delta K$ ,

$$J(t) = \frac{(a-b)^2 w}{(B+Aw)^2} = -\frac{\lambda}{\delta}(I-a)(I-b), \quad (1.42)$$

$$K(t) = \left[1 + \frac{2\lambda}{\delta}(1-I)\right]J = \frac{2\lambda}{\delta}(\kappa-I)J, \quad (1.43)$$

and the cumulative mean release is a quadratic in  $1-I$  alone,

$$V(t) = (1-I)\left[1 + \frac{\lambda}{\delta}(1-I)\right], \quad (1.44)$$

$$V_{\infty} = \frac{a(1-b)}{a-1} = \frac{aB}{A}, \quad \mathbb{E}[\mathcal{K} \mid \text{burst}] = \frac{a}{a-1},$$

in which  $V_{\infty}$  counts the internally cleared cells with their release of zero and the conditional mean removes them. The integral of the survival kernel, which anchors [section 1.5](#), is

$$\mathbb{E}[T_{\text{prod}}] = \int_0^{\infty} I_{\text{fix}}(\alpha) d\alpha = \frac{1}{\lambda} \log \frac{a}{a-1}. \quad (1.45)$$

**Distribution-level facts.** The burst-time density is  $-I'(t) = \delta J(t)$ , defective with total mass  $1-b$ . The burst size, unconditional, obeys

$$\mathbb{P}\{\mathcal{K} = k\} = \frac{\delta}{\lambda} a^{-k}, \quad k \geq 1, \quad (1.46)$$

with the mass  $b$  at “no burst”; conditioned on bursting this becomes  $(a-1)a^{-k}$ , exactly the quasi-stationary distribution of the load [\[31\]](#) — both geometric on  $\{1, 2, \dots\}$  with ratio  $1/a$ . Given that the burst occurs at time  $t$ , its size is a size-biased draw from the state probabilities,  $\mathbb{E}[\mathcal{K} \mid \tau = t] = K(t)/J(t)$ . The second moment of the release, used for the offspring variance [\(1.25\)](#), is

$$\mathbb{E}[W_t^2] = \frac{2(\lambda - \mu)}{\delta} V - \frac{\lambda + \mu}{\delta} I - K + \frac{\lambda + \mu}{\delta} + 1. \quad (1.47)$$

**Multiplicity of infection.** A cell founded by  $k$  particles releases superlinearly more:

$$g_k(t) = \delta K_k(t) = \delta \left( k K I^{k-1} + k(k-1) J^2 I^{k-2} \right), \quad (1.48)$$

with lifetime yield, for  $k \geq 2$ ,

$$V_{\infty}^{(k)} = (1-b^k) + \frac{2\lambda}{\delta} \left[ \frac{1}{k+1} - b^k + \frac{k b^{k+1}}{k+1} \right] + k(k-1) \frac{\lambda}{\delta} \int_1^b (x-a)(x-b) x^{k-2} dx, \quad (1.49)$$

simplifying to  $V_{\infty}^{(k)} = k + \lambda/\delta$  when  $\mu = 0$ . The conditional mean burst  $V_{\infty}^{(k)}/(1-b^k)$  increases with  $k$ : multiply infected cells burst more often, and bigger. The pair  $(I(\alpha)^k - D(\alpha)^k, \delta K_k(\alpha))$  is the MOI- $k$  row of [table 1.4](#); assumption (A4) sets it aside for the population layer.

## 1.B Laplace transforms of the kernels

Write  $s := r/\theta$ . Both transforms reduce, under  $v = e^{-\theta t}$ , to integrals of the form  $\int_0^1 v^{s-1}(1-v)^n(A+Bv)^{-m} dv$ , which are hypergeometric. For the survival kernel, the partial-fraction

decomposition  $(a - b)/[(A + Bv)(a - bv)] = B/(A + Bv) + b/(a - bv)$  and the standard integral  $\int_0^1 v^s/(1 - zv)dv = (s + 1)^{-1} {}_2F_1(1, s + 1; s + 2; z)$  give

$$\widetilde{I}_{\text{fix}}(r) = \frac{1}{\lambda(s + 1)} \left[ \Psi(A, B, s) + \frac{b}{a} {}_2F_1\left(1, s + 1; s + 2; \frac{b}{a}\right) \right], \quad (1.50)$$

where, so that the hypergeometric series always converges,

$$\Psi(A, B, s) = \begin{cases} {}_2F_1(1, s + 1; s + 2; -\frac{A}{B}), & B \geq A, \\ \frac{B}{A} {}_2F_1(1, s + 1; s + 2; -\frac{B}{A}), & A > B, \end{cases} \quad (1.51)$$

the two branches being equal because both sides equal  $(s + 1)B \int_0^1 v^s/(A + Bv)dv$ . For the release kernel the same substitution and termwise integration give

$$\delta \widetilde{K}(r) = \frac{\delta}{\lambda} \left[ \frac{{}_2F_1(1, s; s + 2; z)}{A(s + 1)} + \frac{2 {}_2F_1(2, s; s + 3; z)}{A^2(s + 1)(s + 2)} \right], \quad z = -\frac{B}{A}, \quad (1.52)$$

again with the understanding that for  $B > A$  the right-hand side is evaluated through its analytic continuation, or through the integral representation from which it came. Both reduce correctly at  $s = 0$ : with  ${}_2F_1(1, 1; 2; z) = -\log(1 - z)/z$ , (1.50) gives  $\widetilde{I}_{\text{fix}}(0) = \lambda^{-1} \log(a/(a - 1)) = \mathbb{E}[T_{\text{prod}}]$ , and (1.52) gives  $\delta \widetilde{K}(0) = (\delta/\lambda)(1/A + 1/A^2) = aB/A = V_{\infty}$ , using  $AB = \delta/\lambda$ . Both reductions were verified against direct quadrature to relative error  $\sim 10^{-8}$ . Where the transforms are needed as functions of time, standard numerical inversions apply [32, 33]. The curves of figure 1.5 are computed by direct quadrature.

## 1.C Master formula table

Throughout:  $w = e^{\theta t}$ ,  $\theta = \lambda(a - b)$ ,  $A = a - 1$ ,  $B = 1 - b$ ,  $L = a(1 - b)$ ,  $\kappa = 1 + \delta/(2\lambda)$ .

**Specialisation**  $\mu = 0$ .  $b = 0$ ,  $a = 1 + \delta/\lambda$ ,  $D \equiv 0$ ,  $I_{\text{fix}} = I$ ,  $V_{\infty} = 1 + \lambda/\delta = \mathbb{E}[X]_{\text{QS}}$ ,  $\mathbb{E}[T_{\text{prod}}] = \lambda^{-1} \log(1 + \lambda/\delta)$ ,  $\mathbb{P}\{\mathcal{K} = k\} = s\rho^{k-1}$  with  $s = \delta/(\lambda + \delta)$ ,  $\rho = 1 - s$ ; flooding favours bursting for establishment at every  $\delta > 0$ .

**The boundary**  $L = 1$ .  $\delta = \lambda\mu/(\lambda + \mu)$ ,  $b = \delta/\lambda$ ; the offspring law is geometric with mean  $m$ ;  $z_{\text{ext}}^{\text{burst}} = z_{\text{ext}}^{\text{bud}}$  for every  $q$ ; offspring variances equal.



| Quantity                                              | Formula                                                                                                                                                       |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Renewal system                                        | $\mathcal{I} = \mathcal{I}_0 I_{\text{fix}} + i * I_{\text{fix}}, \quad \mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}, \quad i = \gamma T\mathcal{V}$ |
| Survival kernel                                       | $S(\alpha) = I_{\text{fix}}(\alpha)$                                                                                                                          |
| Release kernel                                        | $g(\alpha) = \delta K(\alpha) = V'(\alpha)$                                                                                                                   |
| Generation kernel                                     | $\mathcal{A}(\alpha) = \gamma T(g * e^{-c \cdot})(\alpha), \quad \tilde{\mathcal{A}}(r) = \gamma T \tilde{g}(r)/(r + c)$                                      |
| $p_{\text{eff}}(r)$                                   | $\delta \tilde{K}(r) / \tilde{I}_{\text{fix}}(r)$                                                                                                             |
| $d_{\mathcal{I}, \text{eff}}(r)$                      | $1 / \tilde{I}_{\text{fix}}(r) - r$                                                                                                                           |
| Limits $r = 0$                                        | $p_{\text{eff}} = \lambda V_{\infty} / \log(a/(a - 1)), \quad d_{\text{eff}} = \lambda / \log(a/(a - 1))$                                                     |
| Limits $r \rightarrow \infty$                         | $p_{\text{eff}} \rightarrow \delta, \quad d_{\text{eff}} \rightarrow \mu + \delta$                                                                            |
| Limit $r \downarrow -\theta$                          | $p_{\text{eff}} \rightarrow \delta \mathbb{E}[X^2]_{\text{QS}} = \delta a(a + 1)/(a - 1)^2$                                                                   |
| Span of the map                                       | $p_{\text{eff}}(-\theta) / p_{\text{eff}}(\infty) = \mathbb{E}[X^2]_{\text{QS}}$                                                                              |
| Characteristic eq.                                    | $r + c = \gamma T \delta \tilde{K}(r), \quad \text{equivalently } \tilde{\mathcal{A}}(r) = 1$                                                                 |
| $R_0$                                                 | $\gamma T V_{\infty} / c = \int_0^{\infty} \mathcal{A}$                                                                                                       |
| Branching mean                                        | $m = q V_{\infty}, \quad q = \gamma T / (\gamma T + c)$                                                                                                       |
| $G_{\text{off}}(z)$                                   | $b + (\delta/\lambda) y / (a - y), \quad y = 1 - q + qz$                                                                                                      |
| $z_{\text{ext}}^{\text{burst}}$                       | $[a(1 - q + qb) - 1 + q] / q$                                                                                                                                 |
| $z_{\text{ext}}^{\text{bud}}$                         | $1/m$ (geometric output, matched mean)                                                                                                                        |
| Flooding parameter                                    | $L = a(1 - b) = a - \mu/\lambda, \quad V_{\infty} = L/(a - 1)$                                                                                                |
| Flooding criterion                                    | $1/\delta < 1/\lambda + 1/\mu \quad (L > 1)$                                                                                                                  |
| $\Delta z_{\text{ext}}$                               | $(L - 1)(1/m - 1)$                                                                                                                                            |
| $\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}}$ | $2q^2 V_{\infty} (L - 1) / (a - 1)$                                                                                                                           |
| $L = 1$                                               | offspring law exactly geometric; variances equal                                                                                                              |
| Growth trade-off                                      | $r_{\text{bud}} > r_{\text{burst}}$ at matched $R_0, \mathbb{E}[T_{\text{prod}}]$                                                                             |

**Table 1.7:** Population-level quantities, in one place.



# Bibliography

- [1] Angela R McLean. The population dynamics of HIV infection. *Journal of Theoretical Biology*, 1993. Details to be verified against the Chapter 1 source.
- [2] Martin A Nowak and Charles R M Bangham. Population dynamics of immune responses to persistent viruses. *Science*, 272(5258):74–79, 1996.
- [3] Alan S Perelson, Avidan U Neumann, Martin Markowitz, John M Leonard, and David D Ho. HIV-1 dynamics in vivo: virion clearance rate, infected cell life-span, and viral generation time. *Science*, 271(5255):1582–1586, 1996.
- [4] Stanca M. Ciupe and Jessica M. Conway. Incorporating intracellular processes in virus dynamics models—a review. *Microorganisms*, 12(5):900, 2024.
- [5] A. V. M. Herz, S. Bonhoeffer, R. M. Anderson, R. M. May, and M. A. Nowak. Viral dynamics in vivo: limitations on estimates of intracellular delay and virus decay. *Proceedings of the National Academy of Sciences USA*, 93(14):7247–7251, 1996.
- [6] John E. Mittler, Bernhard Sulzer, Avidan U. Neumann, and Alan S. Perelson. Influence of delayed viral production on viral dynamics in HIV-1 infected patients. *Mathematical Biosciences*, 152(2):143–163, 1998.
- [7] Narendra M. Dixit and Alan S. Perelson. Complex patterns of viral load decay under antiretroviral therapy: influence of pharmacokinetics and intracellular delay. *Journal of Theoretical Biology*, 226(1):95–109, 2004.
- [8] Yusuke Kakizoe, Shinji Nakaoka, Catherine A. A. Beauchemin, Satoru Morita, Hiromi Mori, Tatsuhiko Igarashi, Kazuyuki Aihara, Tomoyuki Miura, and Shingo Iwami. A method to determine the duration of the eclipse phase for in vitro infection with a highly pathogenic SHIV strain. *Scientific Reports*, 5:10371, 2015.
- [9] Jeremie Guedj, Harel Dahari, Libin Rong, Natasha D. Sansone, Richard E. Nettles, Scott J. Cotler, Thomas J. Layden, Susan L. Uprichard, and Alan S. Perelson. Modeling shows that the NS5A inhibitor daclatasvir has two modes of action and yields a shorter estimate of the hepatitis C virus half-life. *Proceedings of the National Academy of Sciences USA*, 110(10):3991–3996, 2013.
- [10] Libin Rong and Alan S. Perelson. Mathematical analysis of multiscale models for hepatitis C virus dynamics under therapy with direct-acting antiviral agents. *Mathematical Biosciences*, 245(1):22–30, 2013.
- [11] Alun L. Lloyd. The dependence of viral parameter estimates on the assumed viral life cycle: limitations of studies of viral load data. *Proceedings of the Royal Society B*, 268(1469):847–854, 2001.
- [12] James B. Gilmore, Anthony D. Kelleher, David A. Cooper, and John M. Murray. Explaining the determinants of first phase HIV decay dynamics through the effects of stage-dependent drug action. *PLoS Computational Biology*, 9(3):e1002971, 2013.
- [13] Anderson Gray McKendrick. Applications of mathematics to medical problems. *Proceedings of the Edinburgh Mathematical Society*, 44:98–134, 1926.
- [14] Heinz von Foerster. Some remarks on changing populations. In Frank Stohlman, editor, *The Kinetics of Cellular Proliferation*, pages 382–407. Grune & Stratton, New York, 1959.
- [15] J. A. J. Metz and Odo Diekmann. *The Dynamics of Physiologically Structured Populations*, volume 68 of *Lecture Notes in Biomathematics*. Springer, 1986.
- [16] Patrick W. Nelson, Michael A. Gilchrist, Daniel Coombs, James M. Hyman, and Alan S. Perelson. An age-structured model of HIV infection that allows for variations in the production rate of viral particles and the death rate of productively infected cells. *Mathematical Biosciences and Engineering*, 1(2):267–288, 2004.
- [17] Odo Diekmann, J. A. P. Heesterbeek, and J. A. J. Metz. On the definition and the computation of the basic reproduction ratio  $R_0$  in models for infectious diseases in heterogeneous populations. *Journal of Mathematical Biology*, 28(4):365–382, 1990.
- [18] Jacco Wallinga and Marc Lipsitch. How generation intervals shape the relationship between growth rates and reproductive numbers. *Proceedings of the Royal Society B*, 274(1609):599–604, 2007.
- [19] Hongyu Miao, Xiaohua Xia, Alan S. Perelson, and Hulin Wu. On identifiability of nonlinear ODE models and applications in viral dynamics. *SIAM Review*, 53(1):3–39, 2011.

- [20] James E Pearson, Pavel Krapivsky, and Alan S Perelson. Stochastic theory of early viral infection: continuous versus burst production of virions. *PLoS Computational Biology*, 7(2):e1001058, 2011.
- [21] Ing-Nang Wang. Lysis timing and bacteriophage fitness. *Genetics*, 172(1):17–26, 2006.
- [22] Alun L. Lloyd. Realistic distributions of infectious periods in epidemic models: changing patterns of persistence and dynamics. *Theoretical Population Biology*, 60(1):59–71, 2001. TODO(verify).
- [23] David Champredon and Jonathan Dushoff. Intrinsic and realized generation intervals in infectious-disease transmission. *Proceedings of the Royal Society B*, 282(1821):20152026, 2015. TODO(verify).
- [24] Jennifer S. Lord and Michael B. Bonsall. The evolutionary dynamics of viruses: virion release strategies, time delays and fitness minima. *Virus Evolution*, 7(2):veab039, 2021.
- [25] Michael A Gilchrist and Daniel Coombs. Evolution of virulence: interdependence, constraints and selection using nested models. *Theoretical Population Biology*, 69(2):145–153, 2006.
- [26] Jason M Hataye, Joseph P Casazza, Katharine Best, C Jason Liang, Taina T Immonen, Douglas R Ambrozak, Samuel Darko, Amy R Henry, Farida Laboune, Frank Maldarelli, Daniel C Douek, Nicolas W Hengartner, Takuya Yamamoto, Brandon F Keele, Alan S Perelson, and Richard A Koup. Principles governing establishment versus collapse of HIV-1 cellular spread. *Cell Host & Microbe*, 26(6):748–763, 2019.
- [27] Avidan U. Neumann, Nancy P. Lam, Harel Dahari, David R. Gretch, Thelma E. Wiley, Thomas J. Layden, and Alan S. Perelson. Hepatitis C viral dynamics in vivo and the antiviral efficacy of interferon- $\alpha$  therapy. *Science*, 282(5386):103–107, 1998.
- [28] Katia Koelle, Alexander P. Farrell, Christopher B. Brooke, and Ruian Ke. Within-host infectious disease models accommodating cellular coinfection, with an application to influenza. *Virus Evolution*, 5(2):vez018, 2019.
- [29] Jonathan Carruthers, Grant Lythe, Martín López-García, Joseph Gillard, Thomas R Laws, Roman Lukaszewski, and Carmen Molina-París. Stochastic dynamics of *Francisella tularensis* infection and replication. *PLoS Computational Biology*, 16(6):e1007752, 2020.
- [30] Bevelynn Williams, Jonathan Carruthers, Joseph J Gillard, Grant Lythe, Alan S Perelson, Ruy M Ribeiro, Carmen Molina-París, and Martín López-García. The reproduction number and its probability distribution for stochastic viral dynamics. *Journal of the Royal Society Interface*, 21(210):20230400, 2024.
- [31] Erik A van Doorn and Philip K Pollett. Quasi-stationary distributions. University of Twente, 2011. TODO(verify): locate the published version of this survey.
- [32] Harald Stehfest. Algorithm 368: Numerical inversion of Laplace transforms. *Communications of the ACM*, 13(1):47–49, 1970.
- [33] Donald P Gaver. Observations on the numerical inversion of Laplace transforms. *Mathematics of Computation*, 19:131–137, 1965.